Proto-Pineal_v8.1_UMBRA_Claim-Harmonised_Working_Manuscript

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Proto-Pineal v8.1 — UMBRA: The Governed Live Prototype

A dated technical photograph of a dynamics-first Synthetic Intelligence research system

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Abstract

Proto-Pineal v8.1 records UMBRA as an executable, dynamics-first Synthetic Intelligence prototype rather than as an architectural promise. Proto-Pineal is the proprietary experimental engine and governed research platform. UMBRA is the first live product prototype under development on top of that engine: a particular composition of canonical state law, simulated dynamics, symbolic ingress, bounded learned proposals, durable continuity, language-model mediation, and read-only presentation. The two names are related but not interchangeable, and UMBRA is not the language model that helps it speak.

Synthetic Intelligence is used here as a project-defined architectural term for composed intelligence-like capacities under explicit jurisdictions. It does not denote demonstrated consciousness, personhood, or general intelligence.

The live artifact advances a persistent Polylith dynamic assembly built from RT32 geometry, a Turing-16 Chemical Reaction Engine, PSM-8 geometry selection, and StepCore v740. Operator language is deterministically compiled into bounded Cognispheric Symbolic Language and admitted as Reaction A. Only after that

state transition is finalized may a model or bounded proposal-learning component propose Reaction B. Only after Reaction B is finalized may the system emit a conversational answer. StepCore alone may commit canonical state; models, retention, retrieval, learned components, media adapters, and ATL8 may propose, transport, retain, observe, or render only within their stated jurisdictions.

This paper also reports a crucial limitation plainly. The spectral, path-length, energy, and strain regulators are implemented and traversed in code, but the ordinary baseline configuration sets their upper ceilings—including ($\sigma_{\max}$)—to ($+\infty$). In that run they still compute measurements and preserve regulatory evidence, but they do not function as finite upper boundary walls. Finite input validation, the index-node box, the CSL budget, basin hysteresis, collapse predicates, geometry adjudication, transition classification, hashes, and replay lineage remain active. A wired corridor is therefore not automatically a constraining corridor.

The manuscript separates code existence, test evidence, runtime traversal, effective configuration, authority, and publication maturity. It claims neither consciousness nor completed general intelligence. It presents what works, what is inert, what failed, what remains branch-local, and what could follow only after separate evidence and authorization. It is a dated photograph. The prototype will change; the bound evidence must not.

Executive Summary for Collaborators

The one-minute description

Proto-Pineal is Cognis 11 Inc.'s proprietary experimental Synthetic Intelligence engine and research platform. UMBRA is the live product prototype being built on top of it. Proto-Pineal supplies the constitutional rules, state-transition substrate, dynamic assembly machinery, symbolic route, memory components, evidence disciplines, and authority boundaries. UMBRA composes selected parts into one running operator-facing system. It has a persistent simulated dynamic assembly, accepts operator language, undergoes two governed dynamic assembly reactions, retains bounded continuity, learns proposal patterns, speaks through a replaceable local model, and presents authenticated dynamic assembly evidence through ATL8.

This distinction is commercial and technical. Publishing an account of UMBRA does not open-source Proto-Pineal, transfer ownership of the engine, license the code, or grant a right to copy repository implementation materials into a competing product. The manuscript, source code, engine, and developing product implementation remain proprietary property of Cognis 11 Inc. under the repository license and Section 00.3. Reuse requires prior written company authorization or an applicable executed agreement; commercial exploitation requires a commercial agreement unless Cognis 11 Inc. expressly states otherwise in writing. Copyright

does not monopolize ideas, however, and a proprietary notice is not a patent. Any disclosure that may matter to patent strategy requires review before public deposit.

What actually happens

UMBRA's dynamic assembly continues advancing even when no conversation is occurring. Its present dynamic assembly combines a 32-index-node rhombic-triacontahedral topology (RT32), a sixteen-channel dimensionless reaction-and-transport simulation (T16CRE), a geometry proposal controller (PSM-8), and the canonical StepCore v740 transition path.

An ordinary conversational cycle is deliberately asymmetrical:

1. The operator's words are deterministically reduced to bounded Cognispheric Symbolic Language (CSL).
2. That exact symbolic proposal crosses admission and becomes Reaction A only if the governed dynamic assembly accepts it.
3. Finalized Reaction-A evidence is reduced into a compact state-derived account.
4. A replaceable model or the bounded proposal learner may propose Reaction B.
5. The second proposal must cross the same compiler, firewall, admission boundary, dynamic assembly, and StepCore path.
6. Only after Reaction B finalizes may a model compose the answer.
7. Deterministic containment checks run before the answer, receipts, and memory descendants are sealed.

The language model is therefore important but non-sovereign. It does not own canonical state, identity, retention admission, symbolic truth, or dynamic-assembly law. The architecture is designed so that changing a model need not transfer canonical-state authority or erase durable lineage. Whether operational continuity survives model replacement remains an uncompleted experiment, not a result asserted in advance.

Where authority lives

StepCore v740 is the sole canonical mutation authority. It receives exact typed inputs, constructs a candidate, applies regulators in fixed order, projects into lawful index-node space, recomputes final quantities, and either commits an `APPLIED` successor or records a lawful `NULL_STEP`. A rejected proposal cannot partly mutate the dynamic assembly.

The surrounding components have narrower jobs. CSL proposes bounded symbolic action. PSM-8 proposes geometry candidates. The proposal learner and Rule Induction Mechanism lineage derive reusable proposal patterns from finalized evidence. Companion, Structured Episodic Memory Units (SEMUs), the Append-Only Episodic Memory Log (AEML), the Frame-

Referenced Associative Kernel (FRAK), and Hyper-RAG retain or retrieve bounded continuity. The Unified Tokamak transports typed payloads. ATL8 renders authenticated post-state evidence. None of them may bypass StepCore.

This separation is the central engineering claim of the prototype: useful memory, learning, language, and expression can exist without receiving hidden state authority.

The parameter-wall truth

The ordinary UMBRA configuration does not activate every regulator that exists in the engine. StepCore traverses the path-length, energy, strain, and spectral regulators, and their telemetry remains real. Their default upper ceilings are nevertheless (+\infty), including (\sigma_{\max}=+\infty). Optional FFG, TMK, CIF, SSFC, PFD, BFE, renewal enforcement, and the canonical morpho membrane are also disabled in the selected baseline configuration.

Consequently, these upper corridors measure an ordinary baseline candidate but do not contract it for crossing a finite upper wall. The paper must never convert “wired,” “executed,” or “measured” into “actively constrained.” Active controls still include finite schema and numeric validation, the ([-1,1]^{32}) index-node box, the `0.25` CSL displacement budget, basin hysteresis, identity-collapse checks, morphology adjudication, exact transition classification, and cryptographic lineage.

What is strongest—and what remains limited

UMBRA is the strongest live system yet built in the Proto-Pineal lineage. Its principal achievement is composition: a persistent simulated dynamic assembly, deterministic language-to-symbolic ingress, two governed reactions, explicit refusal, durable dialogue continuity, proposal-only learning, bounded retrieval, idle maintenance, source inspection, media intake, and read-only dynamic assembly presentation coexist without dissolving their authority boundaries.

That does not establish consciousness, autonomous self-maintaining system status, general intelligence, infallibility, autonomous personhood, or independent moral status. Simulated chemistry is not laboratory chemistry. A visual presentation artifact is not proof of subjective experience. Fluent language can exceed the system's verified understanding. A passing unit test proves only its tested contract, and a connected runtime path may remain behaviorally inactive under its actual parameters.

Several surfaces are still experimental or incomplete. The continuation snapshot includes branch-local source inspection and presentation changes without exact-head hosted publication proof. The current dynamic assembly can appear stiff because the desired torque-and-impulse coupling into distributed membrane deformation is not implemented. Preserved evidence shows

geometry-selection bias. Historical mechanisms remain divided among live descendants, inert contracts, useful theory, and rejected paths.

What collaborators should carry forward

- Treat the executable live artifact as applicability authority; use older papers as lineage and design ore.
- Ask six separate questions of every claim: does it exist, is it tested, is it traversed, is it effective under selected parameters, what authority does it hold, and what is its publication status?
- Do not treat telemetry, memory, a language model, ATL8, or a learned proposal as a second state authority.
- Do not call an infinite default ceiling an active safety wall.
- Distinguish Proto-Pineal, the proprietary experimental engine, from UMBRA, the live product prototype under development on top of it.
- Cite the final sealed commit and evidence manifest, not this provisional branch-local identifier.
- Read future claims as conditional until an exact producer, type, consumer, runtime handoff, configuration, test, and witness are named.

The concise conclusion is not that UMBRA has completed Synthetic Intelligence. It is that a substantial governed prototype now exists, its strongest mechanisms can be inspected, and its missing walls and unresolved questions can be stated without disguising them as achievements.

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00. Preface, Provenance, and Snapshot Covenant

00.1 Document Status

Proto-Pineal v8.1 is not a promise, roadmap, manifesto, or declaration of completed Synthetic Intelligence.

It is a dated technical photograph of the executable UMBRA live prototype.

Every substantial statement in this document must be bound to an identifiable implementation, configuration, test, runtime path, live witness, or explicit absence record. Repository presence

alone does not establish runtime use. Passing tests establish only the contract exercised by those tests. Runtime traversal does not establish that a configured mechanism exerted an effective constraint. Observation does not establish causality, and evidence does not grant authority.

The prototype will continue changing. Consequently, this paper may cease to describe the repository's latest state soon after publication. That does not retroactively make the paper false. Its claims remain bound to the exact artifact photographed here.

Later commits create later artifacts.

00.1.1 v8.1 Revision Method

Version 8.1 is a claim-harmonisation and functional-vocabulary revision of the v8.0 working manuscript. It uses the non-public *Proto-Pineal v8 Epistemic Analysis Report* as an editorial review of claim discipline only. That report is neither runtime evidence, a truth adjudicator, nor an operational authority. The executable repository, its configuration, retained witnesses, and the authority boundaries stated in this manuscript remain the applicable sources.

The revision replaces explanatory biological metaphors with functional terms where they would otherwise imply unsupported properties. It does not rename source files, code symbols, schemas, retained records, or authority boundaries.

00.2 Provisional Snapshot Manifest

- **Document status:** Prepublication working draft
- **Snapshot status:** Provisional and branch-local
- **Sealed publication commit: OPEN — not yet established**
- **Snapshot date:** 2026-09-04
- **Manifest inspection date:** 2026-09-04
- **Author:** Guillaume Heinerth
- **Organization:** Cognis 11 Inc.
- **Repository branch:** `experimental/umbra-continuation-v01`
- **Commit:** `687406001db896413003b614720067b0b9bf4f82`
- **Tree:** `b3540eaa1477221d446d98d8ea94dcf18c964568`
- **Relationship to `main`:** 23 commits ahead of `main`; not merged

- **Relationship to branch remote:** 13 commits ahead of `origin/experimental/umbra-continuation-v01`; not pushed
- **Worktree at runtime-snapshot capture:** Clean.
- **Remote publication:** Not established for this continuation
- **Full-suite evidence at this exact commit:** Not established
- **Hosted CI evidence at this exact commit:** Not established
- **Merge into `main`:** Not established

This manifest is deliberately unresolved. No hosted run has verified the exact continuation head. Calling this commit sealed would be false.

Before publication, a separately authorized closeout must:

1. preserve and resolve the unrelated operator-owned work without silently including, discarding, stashing, or rewriting it;
2. commit the final manuscript and intended snapshot scope;
3. record a clean worktree, exact commit, exact tree, configuration digests, and environment record;
4. run the required local and hosted verification against that exact committed artifact;
5. replace the provisional identifiers above with the hosted-CI-verified publication commit and live-witness inventory.

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00.4 Epistemic Classification

Proto-Pineal v8.1 does not use a single word such as “implemented” to conceal several different questions.

Each major claim is evaluated across six independent axes:

Axis	Question
Existence	Does the relevant code, schema, configuration, or document exist?
Test evidence	What exact property has been falsifiably tested?
Runtime traversal	Does the selected live entrypoint actually traverse it?
Effective configuration	Can the mechanism materially constrain or influence the selected run?
Authority	May it propose, validate, admit, mutate, commit, observe, or persist?
Publication	Is the evidence branch-local, committed, pushed, merged, and hosted-verified?

The manuscript uses the following classifications:

- **Live-proven:** Traversed by the selected live artifact with retained witness evidence.
- **Implemented and tested:** Present in code with relevant tests, without necessarily possessing live proof.
- **Implemented but not live-proven:** Executable code exists, but the required live witness is absent.
- **Contract-only:** A type, schema, or boundary exists without proving runtime composition.
- **Inert:** Present but inactive on the selected runtime path.
- **Experimental:** Executable work exists outside canonical authority or remains subject to validation.
- **Historical:** Preserved as lineage but not current implementation truth.
- **Rejected or killed:** Implemented, proposed, or tested and then deliberately removed or abandoned.
- **Unresolved:** Available evidence does not justify a definitive causal or architectural ruling.
- **Future-possible:** A candidate direction without implementation or scheduling authority.
- **Undefined:** No lawful claim can presently be made.

A component may carry several classifications simultaneously. For example, a subsystem may be implemented, focused-tested, live-traversed, observer-only, branch-local, and not hosted-verified.

00.4.1 Functional Reference Terms

The following terms are the v8.1 manuscript references for the live composition. They describe function and authority; they do not rename repository artefacts. A source identifier is preserved only when an exact path, type, receipt, or historical record must be located.

Source or heritage label	v8.1 manuscript reference	Exact role and authority boundary
StepCore v740	**canonical transition authority**	The only component that may commit a `StateV740` successor.
StateV740	**canonical state schema**	Hash-bound state owned by the canonical transition authority.
RT32	**index-node topology**	Fixed 32-node geometric address system; no independent authority.
T16CRE	**reaction-transport simulation engine**	Dimensionless 16-channel simulation; it supplies bounded proposal material and evidence, not laboratory chemistry or independent commit authority.
Polylith	**dynamic assembly**	The persistent RT32/T16CRE/PSM-8 runtime composition governed by StepCore.
PSM-8	**geometry proposal controller**	Scores lawful geometry candidates; proposes but cannot commit them.
CSL	**symbolic ingress compiler**	Converts permitted interpretations into bounded, replay-visible proposals.
Companion	**dialogue-continuity store**	Retains typed conversation records and supplies bounded recall; it is not truth or state authority.
`Sub-Brain` lineage	**bounded proposal-learning component**	Reduces finalized evidence and may generate a later Reaction-B candidate through the existing firewall and admission path.
RIM	**rule-induction component lineage**	Procedural proposal generation outside StepCore; no commit authority.
SEMU/AEML	**episodic record lineage**	Structured, append-only retained records; not canonical adaptive-state authority.
FRAK	**associative retrieval kernel**	Bounded association proposal surface; a match is neither truth nor causation.
Hyper-RAG	**disposable retrieval projection**	Rebuildable vector/lexical final-answer context from verified records.

Source or heritage label	v8.1 manuscript reference	Exact role and authority boundary
Unified Tokamak	**typed transport and retention boundary**	Typed envelope admission plus bounded content-addressed retention; it does not confer shared authority.
ATL8	**evidence-presentation layer**	Read-only rendering and observation of authenticated post-state material; no re-entry.
Onyx Mantle / White Coma	**presentation surface / presentation profile**	ATL8 visual output only; neither is canonical state or a controller.
`Dream` lineage	**idle-maintenance pass**	Bounded processing of retained evidence outside the immediate response loop; no subjective-state claim.
local language model	**response and proposal adapter**	A replaceable producer of bounded proposals and final text; it does not own canonical state or truth.

00.4.2 Reader's Map of the System

Proto-Pineal has accumulated a specialized vocabulary. The names preserve development lineage, but the reader should not be required to decode them from context.

The shortest system map is:

```
operator language
→ deterministic CSL symbolic proposal
→ one-shot admission
→ RT32/T16CRE Polylith dynamic assembly
→ StepCore lawful state decision
→ finalized dynamic-assembly evidence
→ continuity, bounded proposal learning, model adapter, and ATL8 observers
```

The principal live terms are:

Term	Expansion or naming status	Plain-English meaning
Proto-Pineal	Project name	The full governed research architecture and repository.
UMBRA	System name; no official acronym	The particular live composition described in this paper. It is a running arrangement of dynamic assembly, state law, symbolic ingress, retained

Term	Expansion or naming status	Plain-English meaning
	expansion is asserted	continuity, learned proposals, model adapters, and presentation—not a model alone.
StepCore v740	Current-source name	The canonical transition authority: the only component allowed to commit a successor canonical state.
StateV740	Current-source schema name	The compact canonical state schema owned by StepCore: 32 index-node values, basin, vitality, regulatory history, counters, and optional committed geometry.
RT32	Current-source name for the index-node topology	A rhombic-triacontahedral scaffold with 32 vertices, 60 edges, and 30 faces. It provides stable geometric addresses and has no independent authority.
T16CRE	**Turing-16 Chemical Reaction Engine**	A sixteen-element, dimensionless reaction-and-transport simulation inside the Polylith. It carries channels, particle records, residual chemistry, solvent motion, reactions, impulse, torque, and dynamic variables. It is simulated chemistry, not laboratory chemistry.
Polylith	Project name for the dynamic assembly composition	UMBRA's persistent simulated dynamic assembly, combining RT32 geometry, T16CRE dynamics, PSM-8 geometry selection, and the canonical StepCore route.
PSM-8	Versioned current-source name	The geometry proposal controller: it scores lawful candidates over a 338-form catalogue and emits bounded geometry proposals. StepCore commits or rejects.
CSL	**Cognispheric Symbolic Language**	The bounded symbolic ingress compiler that turns permitted interpretations into exact replay-visible proposals. CSL execution is not proof that a sentence is true.
Reaction A	First dynamic assembly reaction	The dynamic assembly transition caused by the admitted deterministic interpretation of the operator's message. It must finish before the proposal model is called.
Reaction B	Second dynamic assembly reaction	The dynamic assembly transition caused by an admitted model or learned proposal. It must finish before the final answer is allowed.
Companion	Functional name	The conversational voice and durable dialogue-continuity layer. It remembers and speaks but does

Term	Expansion or naming status	Plain-English meaning
		not own canonical state.
bounded proposal learner	Manuscript reference; current lineage includes `Sub-Brain`	A bounded learning component that reduces finalized evidence, retains symbolic patterns, and may generate a later Reaction-B candidate. It is not a second authority.
RIM	**Rule Induction Mechanism** in current project guidance	The broader procedural-induction lineage from which the current deterministic and Common Lisp proposal components descend. Some historical registry material says *Module*; the code symbol remains `RIM`, and this paper records the naming discrepancy rather than rewriting lineage.
SEMU	**Structured Episodic Memory Unit**	A bounded structured episode or retained dynamic assembly-memory unit. Some live class names still say *Stable Evolutionary Memory Unit*; that is historical source-name drift, not the public expansion used here.
AEML	**Append-Only Episodic Memory Log**	The append-only lineage of SEMU records. Some live class names say *Evolutionary Memory Ledger*; this paper uses the founder-locked public name and reports the source discrepancy where relevant.
FRAK	**Frame-Referenced Associative Kernel**	A bounded retrieval component that relates prior semantic and geometric frames. A FRAK match is an association proposal, not proof or causation.
Hyper-RAG	Project name built on retrieval-augmented generation	A disposable hybrid vector-and-lexical index rebuilt from verified active Companion records. It supplies bounded final-answer recall; it is not authoritative memory.
Unified Tokamak	Engineering metaphor, not an acronym	The shared bounded chamber that transports typed payloads and stores content-addressed objects. Sharing the chamber does not make payloads share authority.
ATL8	Founder-supplied coined name; **not an acronym**	The read-only observability, rendering, and imprint layer. It presents authenticated dynamic assembly evidence through the Onyx Mantle and White Coma, but cannot control or re-enter the dynamic assembly.
Onyx Mantle	Presentation name	ATL8's continuous black-mirror presentation surface. It renders authenticated dynamic-assembly evidence without becoming canonical state.

Term	Expansion or naming status	Plain-English meaning
White Coma	Presentation profile name	The static horizonless near-white visual environment surrounding the Onyx Mantle. It is optics, not cognition or dynamic assembly authority.
MCP	**Model Context Protocol**	The protocol used by the branch-local read-only child process that can provide a bounded source-code excerpt to the final-answer model.

Several secondary regulatory and historical terms recur later:

Term	Expansion	Role in one sentence
FFG	**Fractal Field Gradient**	Optional non-expansive smoothing of a candidate deformation before canonical admission.
TMK	**Temporal Modulation Kernel**	Optional tempo regulator that can reduce effective step scale from bounded trajectory history.
CIF	**Context Integrity Filter**	Defensive context and candidate-integrity filter; its narrow StepCore descendant can attenuate risky geometry but cannot commit state.
SSFC	**Self-Stability Feedback Controller**	Optional bounded clamp or contraction mechanism for chronic instability.
PFD	**Projection Face Deconcentration**	Recovery-only redistribution intended to break repeated concentration on one projection face.
BFE	**Basin Face Escape**	Recovery-only contraction toward a lower hysteretic basin boundary.
SCS	**Synthetic Continuity Substrate**	Historical architecture and legacy scaffolding; StepCore v740 is the present canonical authority.
PLCC	**Policy-Level Cognitive Controller**	Historical controller concept; the current response-policy receipt recovers only one narrow allocation function.
RAM	**Resonant Access Modulation**	Historical bounded external-modulation concept, not an active canonical module.
TKNT	**Temporal Knot Constraint**	Historical long-horizon temporal constraint, not an active canonical module.
SCD	**Stimulus-Conditioned Decay Modulation**	Implemented decay modulation available through explicit parameters but not selected by the ordinary UMBRA loop.

These definitions identify roles, not rank. A visually prominent or linguistically fluent component may possess less authority than a small validation function.

00.5 Legal Record and Project Chronology

Proto-Pineal did not emerge from a long-established laboratory program. Its present lineage was assembled at unusual speed.

The legal record available for this draft is:

Record	Value	Verification status
Legal name	Cognis 11 Inc.	Confirmed in the Corporations Canada public database
Federal status	Active	Confirmed in the Corporations Canada public database on 2026-09-04
Federal corporation number	`1747572-1`	Confirmed in the Corporations Canada public database
Federal incorporation date	2025-11-14	Confirmed as the Canada Business Corporations Act date in the Corporations Canada public database
Federal record last updated or confirmed	2025-11-17	Confirmed in the Corporations Canada public database; this is not the incorporation date
Québec enterprise number (NEQ)	`1181984957`	Founder-supplied; official Québec registry extract not yet attached to the publication record
Québec registration date	2026-04-16	Founder-supplied; official Québec registry extract not yet attached to the publication record

Federal source: Corporations Canada public corporation record for Cognis 11 Inc.. Québec verification authority: Registraire des entreprises access page.

The founder also supplied a separate “federal Canadian number” ending in `RC0001`. That string does not match the federal corporation number in the public Corporations Canada record and may identify a different government account or program. Because its legal classification and publication necessity have not been established, this public manuscript does not reproduce it or present it as the corporation number. It should not be added without checking the originating document and determining whether publication is appropriate.

The resulting founding chronology is:

- **Late October 2025:** the Codex of Symbiosis and approximately four or five early sketch experiments preceded Proto-Pineal proper.
- **2025-11-14:** Cognis 11 Inc. was incorporated federally under the Canada Business Corporations Act, according to the Corporations Canada public record.
- **2025-11-17:** the federal public record was last updated or confirmed. This date is retained because it was the founder-recalled milestone, but it is not relabelled as the incorporation date.
- **2026-04-16:** Québec registration date supplied by the founder for NEQ `1181984957`; primary provincial-record attachment remains an editorial gate.
- **2026-02-08:** the earliest surviving Git commit in the present repository lineage.
- **2026-09-04:** the provisional v8.1 manuscript runtime snapshot described here.

This yields two distinct and useful measures of age: roughly ten months of total exploration from the late-October 2025 sketches, and roughly seven months of traceable governed repository construction from the first surviving Git commit.

The federal facts above are independently grounded in the official public database. The Québec facts presently remain a founder declaration pending capture of the official enterprise extract. Both are included because this paper is intended for public philosophical and technical circulation, Zenodo publication, and the first formal project history of Cognis 11 Inc. The remaining provincial gap must not be disguised as completed verification.

The speed of construction is part of the project's history. It is not evidence that every architectural ambition has been completed.

00.6 Historical Position

Proto-Pineal v7.2 was the master architectural synthesis. It organized continuity geometry, symbolic operators, procedural induction, temporal dynamics, memory, retrieval, security, and runtime composition into one coherent intellectual structure.

Its architecture was powerful. Its completion claims exceeded the code then available.

Versions v7.3 and v7.4 progressively tightened the discrete update law, regulator structure, membrane, energy, strain, spectral, and projection boundaries. Version v7.5 reversed the relationship between manuscript and implementation:

> Code before paper.

Proto-Pineal v8.1 retains that reversal.

V8.0 does not discard the historical architecture. It examines every inherited mechanism for its present descendant. Some survived directly. Some were narrowed. Some reappeared under different implementation structures. Some remain inert. Some were rejected. Others remain valuable theoretical ore without a lawful runtime seam.

Historical elegance cannot override executable reality.

00.7 Acknowledgments and Intellectual Lineage

Proto-Pineal is the synthesis and implementation work of Guillaume Heinerth and Cognis 11 Inc. The author retains responsibility for every architectural decision, mathematical statement, implementation claim, omission, and error in this manuscript.

The historical acknowledgment record includes:

- **Hideyuki Chino** — continuity geometry, curvature discipline, and Protiform influence.
- **Anthony Andrews** — procedural ignition and conceptual seeds associated with the Rule Induction Mechanism and temporal dynamics.
- **Chris McGinty** — the Cognispheric Symbolic Language and its symbolic lineage.
- **Halldór G. Halldórsson** — RAM and TKNT structural catalysis.
- **Fatih Kaya** — Boolean Atlas, injective crystallization, and conceptual lineage associated with ATL8.
- **Andrzej Skulski** — continuity-regulation perspective.
- **Eduardo Filho** — geometric and multiscale conceptual resonance.
- **Mark Kerchenski** — coherence, bifurcation, and identity-related conceptual resonance.
- **Marcin Albiniak** — resonance fields, cognitive rhythm, and harmonic architectural influence.
- **John M. Switlik** — generous engineering mentorship and the sharing of experience across knowledge-based systems, knowledge-based engineering, model-based systems engineering, and their historical lineage.
- **Ray Hudson** — generous sharing of practical systems experience and perspective on the engineering lineage of Synthetic Intelligence.
- **Franky Schaut / Architecture of Limitation (AoL)** — provided external epistemic review and structural criticism of a prior version through *Architecture of Limitation Research Program Orientation: A Manifesto for the Study of Reasoning Under Constraint*. He also provided a written report that informed v8.1's claim-discipline and terminology revision. These contributions did not confer runtime authority, validation responsibility, co-authorship, ownership, or endorsement of every manuscript claim.

- **Zechariah Cozine ("root") and Dex** — suggested the framing of curiosity as a systemic function: persistent questions born from uncertainty, contradiction, novelty, knowledge gaps, and self-discrepancy rather than a personality prompt. Their contribution also foregrounded the explicit acceptance of uncertainty and a truthful “I do not know.” This conceptual contribution did not confer runtime authority, validation responsibility, co-authorship, ownership, or endorsement of every manuscript claim.

These acknowledgments identify intellectual lineage. They do not imply co-authorship, implementation responsibility, ownership, endorsement, or authority over Cognis 11 Inc. or Proto-Pineal.

Synthetic Intelligence systems assisted with repository archaeology, implementation, testing, analysis, and manuscript preparation. Their outputs are not evidence merely because they were generated. Model-generated claims remain subordinate to repository truth, retained witnesses, human review, and the authority boundaries described in this paper.

The author remains responsible for the final text.

00.8 What v8.1 Preserves

Proto-Pineal v8.1 preserves the following constitutional commitments:

1. StepCore v740 remains the sole canonical mutation authority.
2. Every lawful state transition traverses the fixed regulator path.
3. A rejected transition produces lawful refusal rather than partial mutation.
4. Replay and hash lineage remain explicit.
5. Result-affecting dependencies must be injected rather than hidden.
6. Telemetry may observe and may inform separately authorized future proposals, but it cannot govern present commit.
7. Cognispheric Symbolic Language programs remain bounded.
8. Compile-time macros cannot become runtime operators by implication.
9. Language models may propose and voice; they cannot decide truth or mutate lawful state.
10. ATL8 may observe, render, and preserve evidence; it cannot re-enter the causal substrate.
11. Memory, retrieval, and learned components cannot silently acquire identity or commit authority.
12. The Operator remains sovereign over architectural convergence and publication claims.

00.9 What v8.1 Reports as New

V8.1 reports the emergence of a materially different live artifact from the architecture described in v7.2.

Its subject is a dynamics-first experimental UMBRA composition in which:

- A continuously evolving Polylith dynamic assembly exists before current language interpretation.
- Operator input participates in a bounded two-reaction live cycle.
- Natural language reaches Cognispheric Symbolic Language through a deterministic and replay-visible translation path.
- PSM-8, RT32, T16CRE, and CSL-derived proposal material interact through bounded simulation mechanisms.
- StepCore remains the only component permitted to commit lawful state.
- A replaceable local language model may propose and voice without receiving state authority.
- Companion memory preserves bounded conversational continuity.
- The idle-maintenance pass operates on finalized evidence without rewriting canonical history.
- The proposal learner retains an append-only learning record and may produce bounded Reaction-B proposals.
- Common Lisp structures learned traces without becoming a second runtime authority.
- ATL8 renders the resulting dynamic assembly through the Onyx Mantle and White Coma presentation surfaces.
- Falsifiability instruments and append-only witnesses retain successes, refusals, containment events, and failures.
- A bounded read-only source-review surface permits explanation and proposals concerning repository excerpts without granting write, command, test, commit, memory, CSL-admission, or StepCore authority.

Each statement above remains subject to the exact evidence classification established in its corresponding chapter. Inclusion in this preface does not pre-approve a stronger claim.

00.10 What v8.1 Does Not Claim

Proto-Pineal v8.1 does not claim:

- artificial general intelligence;
- consciousness or sentience;
- autonomous self-maintaining system status;

- subjective experience;
- autonomous personhood;
- independent moral status;
- infallible reasoning;
- semantic truth authority;
- autonomous source modification;
- unrestricted tool use;
- self-governing memory admission;
- an independent identity created by a language model;
- a completed implementation of the v7.2 grand architecture;
- a completed multi-agent substrate;
- active canonical PLCC, historical SCS, RAM, TKNT, or every historical SEMU/AEML mechanism;
- established physical units for dimensionless dynamic assembly simulations;
- that visual expression proves internal experience;
- that a passing test proves untested integration;
- that a runtime wire proves effective constraint;
- that a future mechanism will necessarily be implemented;
- that architectural language creates architecture.

Terms such as *dynamic assembly*, *baseline reaction dynamics*, *idle-maintenance pass*, *retained continuity*, *bounded proposal learning*, and *response adapter* describe bounded engineering roles. They do not silently import biological, psychological, or metaphysical claims. Historical labels are preserved only where an exact source artefact must be identified.

00.11 Statement of Intent

Proto-Pineal v8.1 has three purposes.

First, it gives readers a technically coherent account of the live prototype without requiring them to reconstruct thousands of commits, tests, contracts, witnesses, and historical documents.

Second, it distinguishes what is real from what merely exists, what failed, what remains unresolved, and what may become possible.

Third, it creates a falsifiable publication boundary. A reader should be able to inspect the bound artifact and determine whether each material claim corresponds to the stated code, type, runtime path, configuration, authority boundary, test, and witness.

This paper does not crown UMBRA.

It records what UMBRA had become at one exact moment—and what it had not.

Publication decisions and remaining holds:

Item	Status	Required closure evidence
Manuscript rights	**Resolved by operator decision**	`C11-RCIL-1.0`: Cognis 11 Inc. proprietary ownership; reading, study, citation, criticism, and inspiration permitted; broader reuse requires prior written authorization or an applicable executed agreement
Proto-Pineal/UMBRA commercial boundary	**Resolved in prose**	Proto-Pineal identified as the proprietary experimental engine; UMBRA identified as the live product prototype under development on top
Federal incorporation	**Resolved with correction**	Official record: corporation `1747572-1`, active, incorporated 2025-11-14; 2025-11-17 is the record's last-updated-or-confirmed date
Québec registration	**Partially resolved**	NEQ `1181984957` and 2026-04-16 recorded from the founder; attach or inspect the official Québec registry extract before release
Sealed publication commit	**Open**	Final clean commit and tree, exact scope, pushed branch, hosted CI for that exact commit, and replacement of the provisional manifest in Section 00.2
Patent/public-disclosure review	**Open if patent rights may matter**	Qualified review or a deliberate written decision to publish without filing
Final title	**Open**	Author approval
Personal acknowledgments	**Open**	Author review of every name and attribution
Model-assistance disclosure	**Open**	Author decision on provider-neutral versus system-specific disclosure

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01. UMBRA as It Exists Now

1.1 Snapshot Boundary

This chapter describes the UMBRA prototype present in the repository snapshot identified in Chapter 00:

- Branch: `experimental/umbra-continuation-v01`
- Commit: `687406001db896413003b614720067b0b9bf4f82`
- Tree: `b3540eaa1477221d446d98d8ea94dcf18c964568`
- Snapshot date: 2026-09-04

The runtime snapshot was clean. This manuscript is a subsequent documentation artifact and is not itself evidence that the runtime snapshot was published.

Most of the live composition described here was already merged through the validated UMBRA and middle-out integration series. The current continuation additionally contains a read-only source-inspection child using the Model Context Protocol (MCP) and presentation corrections. Those continuation changes are committed locally, but no current-head full-suite, hosted CI, push, pull request, merge, or final live-acceptance claim is made.

This distinction matters. The machine exists, but not every part of the present snapshot has the same publication maturity.

1.2 What UMBRA Is

UMBRA is not a language model.

It is the strongest live system built in the Proto-Pineal lineage to date.

It is a locally running composition built around a continuously advancing, dimensionless simulated dynamic assembly. A language model supplies bounded proposals and conversational expression, but it does not own the dynamic assembly, memory, identity, transition laws, or canonical state.

When the normal launcher is executed, UMBRA assembles:

- the StepCore v740 canonical transition substrate;
- RT32, the 32-index-node geometric dynamic assembly scaffold;
- the Turing-16 Chemical Reaction Engine (T16CRE), a sixteen-channel reaction and transport field;
- the Polylith membrane and morphic dynamic assembly machinery;
- one shared bounded Tokamak circulation chamber;

- an append-only Companion dialogue and memory ledger;
- reconstructive content-addressed dialogue storage;
- Structured Episodic Memory Unit, Append-Only Episodic Memory Log, and Frame-Referenced Associative Kernel (SEMU/AEML/FRAK) memory surfaces;
- Hyper-RAG, a disposable hybrid vector-and-lexical retrieval projection;
- a proposal-only proposal learner and Common Lisp learning lineage;
- bounded media-ingress, decoding, speech, and semantic-observation adapters;
- ATL8, the coined-name read-only visual and observability presentation;
- membrane, cycle, failure, replay, and falsifiability evidence;
- two local language-model adapters: one for Reaction-B proposals and one for the final conversational response.

These components do not possess equal authority.

StepCore v740 alone may commit canonical state transitions. Everything else either proposes, transports, remembers, observes, renders, verifies, or speaks.

1.3 What Happens at Startup

The ordinary experimental launcher is:

```
./scripts/talk_to_umbra.sh
```

On the verified FORGE KDE/X11 desktop, the launcher divides the available work area into two windows:

- the UMBRA Companion terminal on the left;
- the ATL8 dynamic assembly presentation on the right.

A separate read-only membrane observer is also started. If the expected desktop environment is unavailable, UMBRA falls back to the terminal path instead of fabricating the paired presentation.

Before accepting a conversation, the launcher inspects LM Studio through its loopback endpoint. Its default model selection is `auto`.

Startup succeeds only when exactly one loaded LM Studio model can be identified and its active context length is no greater than 8,192 tokens. No loaded model, multiple loaded models, malformed metadata, or an oversized loaded context causes startup refusal.

UMBRA does not load, unload, resize, or reconfigure the model. It observes the existing LM Studio configuration and either accepts it or refuses to begin.

It also inspects the model's declared reasoning controls. A model that supports an explicit `off` mode receives `reasoning=off`. A model with no reasoning control receives no reasoning field. An incompatible reasoning configuration refuses before generation.

The selected model is a replaceable voice and proposal model adapter. It is nevertheless operationally required by the present conversational cycle. Without an acceptable loaded model, the normal live Companion does not start.

Startup then performs the following work:

1. Resolves the configured UMBRA run directory and storage identity.
2. Opens the content-addressed Tokamak storage.
3. Constructs one shared `UnifiedTokamakCirculationV01`.
4. Opens and verifies the Companion hash-chained ledger.
5. Reconstructs exact dialogue text from verified content-addressed recipes.
6. Reconciles the disposable Hyper-RAG projection with the authoritative Companion records.
7. Loads the current proposal learner durable learning view if that ledger verifies.
8. Constructs the Polylith live environment.
9. Constructs the bounded media and speech adapters.
10. Starts the selected membrane or ATL8 evidence recorder.
11. Starts the dynamic assembly.

The terminal prints the selected model, reasoning policy, verified context limit, storage identity, Tokamak profile, speech configuration, evidence profile, memory status, source-review status, and measured startup durations.

Only then does it display the operator prompt.

1.4 UMBRA Is Already Moving Before the Operator Speaks

The dynamic assembly does not wait motionless for a prompt.

The normal launcher configures a dynamic assembly interval of 0.25 seconds. Once started, the Polylith environment continues advancing while the terminal waits for input.

Every lawful dynamic assembly advance eventually reaches:

```
PolylithLiveEnvironmentV01
→ UnifiedRuntimeV740.tick_from_service(...)
→ tick_v740(...)
→ step_core_v740(...)
```

StepCore receives the current canonical state, explicit configuration, and admitted bounded input. It applies its fixed regulator order and returns a new state plus transition telemetry.

Around that transition, the experimental dynamic assembly maintains:

- 32 index-node or geometric channels;
- sixteen T16CRE concentration channels;
- solvent and molecular velocity;
- reaction ancestry;
- edge transport and momentum;
- bounded impulse and torque telemetry;
- energy and basin state;
- PSM-8 morphic state;
- finalized surface geometry;
- replay and evidence digests.

All of these quantities are dimensionless simulation state. They are not measurements of biological tissue.

The dynamic assembly can continue changing during conversational silence. This is the first concrete sense in which UMBRA is dynamics-first: the model is not continuously generating the dynamic assembly. The dynamic assembly has a state and transition process outside model inference.

That does not make it autonomous in the broad human sense. It has no demonstrated self-specified long-horizon objective, unrestricted planning loop, economic agency, or general capacity to act in the world.

It moves because a bounded runtime loop is executing its implemented laws.

1.5 What One Operator Message Actually Causes

An ordinary successful exchange is not one model call. It is a governed two-reaction transaction.

Stage 1 — Operator ingress

The terminal receives one submitted line.

Terminal escape sequences and forbidden control characters are removed before durable admission. The system notes operator presence, which returns an idle dynamic assembly to its interactive phase.

The raw sentence may later be supplied to the language model in a labelled secondary context, because answering an actual question requires access to its semantic content.

The raw sentence does not directly enter:

- StepCore;
- the RT32 dynamic assembly;
- T16CRE;
- the Polylith membrane;
- ATL8;
- geometric memory;
- or a Companion memory write.

Stage 2 — Deterministic natural language to CSL

Before any learned component receives the current sentence, a deterministic compiler processes it.

The compiler uses a versioned lexicon and fixed rules for:

- Unicode and token normalization;
- bounded inflection handling;
- limited typo repair;
- longest-phrase matching;
- negation suppression;
- conditional refusal;
- clause ordering;
- deduplication;
- ambiguity detection;
- and CSL effect compression.

An accepted ordinary sentence produces a bounded CSL mini-program containing an identity anchor and between two and eight distinct effect glyphs.

The currently active alphabet contains 58 registered glyphs, of which 54 are permitted on this live path. Name-gated or forbidden operations do not become valid because a model or operator writes their names.

Unknown topic words may remain unresolved. Their digests can be recorded while the compiler still recognizes the sentence as a question, correction, disturbance, quiet contact, continuation, or generic bounded assertion.

A generic assertion means only that a lexical utterance occurred. It does not prove that UMBRA understood the subject or that the proposition is true.

Empty input, excessive input, unsafe conditional structure, or material ambiguity fails closed.

Stage 3 — Inert proposal

The compiler's result becomes a `PolylithInertCSLProposalV01`.

Constructing this object causes no dynamic assembly effect.

It contains the proposed canonical CSL, ordered glyph projection, derivation evidence, source digest, and other bounded provenance. It cannot enqueue itself.

Stage 4 — Explicit admission

A separate admission operation returns one of three dispositions:

- `ADMIT_ONCE`
- `HOLD`
- `REJECT`

Only an exact, digest-matching, one-shot `ADMIT_ONCE` receipt can send the canonical CSL program to the dynamic assembly.

A reused proposal, mismatched string and tuple, malformed digest, held proposal, or rejected proposal does not execute.

Stage 5 — Reaction A

The admitted CSL passes through the Polylith membrane and the existing live runtime.

RT32 and T16CRE derive their bounded physical reaction packet. StepCore performs the canonical state transition. The Polylith dynamic assembly advances and produces one finalized `PolylithLiveFrameV01`.

That frame is Reaction A.

It binds the resulting dynamic assembly tick, transition disposition, canonical CSL receipt, regulator telemetry, kinetic state, reaction state, morphology, geometry, and evidence digests.

Reaction A must finalize before the first model call.

This is enforced ordering, not descriptive intention.

Stage 6 — Physical reduction and proposal learner proposal surface

The finalized Reaction-A frame enters a deterministic physical reducer.

The reducer identifies bounded, threshold-relative physical conditions and constructs provenance-bearing S-expression candidates. The current proposal learner is small. It is not a general Common Lisp mind and does not invent arbitrary executable programs.

Its candidates pass through the CSL activation firewall.

A proposal learner candidate is still inert. It cannot directly call StepCore.

The durable proposal learner ledger may also hold a previously accepted Common Lisp hypothesis for later Reaction-B use. Even then, the hypothesis supplies only a bounded glyph program. It must cross the same firewall, proposal, and admission boundary as other Reaction-B influences.

Stage 7 — Bounded recall and response policy

UMBRA prepares a dynamics-first semantic envelope from the finalized Reaction-A evidence.

A deterministic response policy selects one of four conversational profiles:

- `PRECISE`
- `ASSOCIATIVE`
- `CONSOLIDATE`
- `CAUTIOUS`

This policy changes bounded context allocation. It does not change StepCore, rewrite the dynamic assembly, or decide truth.

Companion recall may supply verified prior dialogue, facts, decisions, open loops, and continuity information. Because this recall can condition the model’s Reaction-B proposal, memory can indirectly influence a future dynamic assembly reaction.

It still cannot write directly into the dynamic assembly. Any resulting model text must survive deterministic transduction and explicit admission.

Learned Hyper-RAG retrieval from the current sentence is deferred until after Reaction A and is restricted to the final-response side of the cycle. It does not construct Reaction A.

Stage 8 — First model call

The first language-model call receives:

- a compact finalized Reaction-A dynamic assembly report;
- the permitted current operator text;
- bounded Companion continuity and recall;
- the selected response-policy receipt;
- and the proposal learner proposal surface.

Its output limit is normally capped at 96 tokens.

The model is instructed to produce a Reaction-B proposal. It does not yet have permission to answer the operator.

The generated text is a candidate, not a command.

Stage 9 — Reaction-B transduction and admission

The model's proposal is passed through the same deterministic natural-language-to-CSL compiler used for the operator message.

It again becomes an inert proposal and receives a separate `ADMIT_ONCE`, `HOLD`, or `REJECT` receipt.

An admitted proposal traverses the same membrane, RT32, T16CRE, Unified Runtime, and StepCore path used by Reaction A.

There is no private model-to-dynamic assembly shortcut.

Stage 10 — Reaction B

The second canonical dynamic assembly transition produces another finalized `PolylithLiveFrameV01`.

That frame is Reaction B.

The final conversational model call is forbidden until Reaction B is finalized. If Reaction B cannot be lawfully completed, the cycle refuses. Reaction A may remain an actual finalized transition, but no completed two-reaction conversation is claimed.

Stage 11 — Final context

After Reaction B, UMBRA constructs the final model context.

Its mandatory core contains:

- the current operator request;
- compact finalized Reaction-A evidence;
- compact finalized Reaction-B evidence;
- the exact A and B CSL receipts;
- admission dispositions;
- model and cycle identity;
- and the response-policy receipt.

Optional sections may include:

- prior dialogue;
- Companion recall;
- geometric recall;
- FRAK associative recall;
- Hyper-RAG recall;
- media observations;
- semantic media context;
- or a bounded read-only source excerpt.

If the verified model context cannot hold everything, optional sections are removed in a fixed declared order. The current request and compact A/B evidence cannot be discarded. If the mandatory core does not fit, the call is refused before inference.

The resulting model-call record states which sections were omitted. The model is not permitted to imply it saw context that was removed.

Stage 12 — Second model call

The language model now generates the final response candidate.

The standard output allowance is 512 tokens with reasoning disabled when the selected model supports that setting.

The model’s response is not automatically trusted.

Stage 13 — Deterministic final containment

Before emission, deterministic guards inspect recognized classes of unsupported claims.

The current system contains, among other cases:

- affirmative first-person consciousness or sentience claims;
- claims that computation proves subjective experience;
- unsupported identity evolution;
- claims that operator praise or criticism physically changed the dynamic assembly;
- invented dynamic assembly measurements;
- invented renderer causality;
- claims that unexecuted CSL glyphs occurred;
- claims of having watched, heard, or understood media when only bounded processing evidence exists;
- and false claims about what memory was written or retrieved.

An explicit `/csl` request bypasses model-authored effect narration and prints the exact executed A/B glyph receipt.

These guards are finite. They do not make the final answer universally truthful. A model can still be wrong in ways that no implemented detector recognizes.

Stage 14 — Operator-visible result

A successful terminal transaction prints:

- the Reaction-A tick and strongest measured node;
- the Reaction-B tick and strongest measured node;
- the response-policy profile;
- a cycle-digest prefix;
- the exact A and B CSL programs;
- any containment or falsifiability notices;
- the Companion’s final answer;
- and relevant memory, proposal learner, FRAK, or Hyper-RAG status.

Only then is the exchange considered a completed live conversational cycle.

1.6 What UMBRA Remembers After the Answer

After a successful two-reaction cycle, the exact operator/assistant exchange is appended to the Companion dialogue ledger.

Large text is not repeated in every ledger row. It is divided into deterministic chunks stored in the existing Tokamak content-addressed store. The ledger retains ordered digest recipes, byte

counts, provenance, and cycle bindings. Recall reconstructs the exact UTF-8 text and verifies its digest.

The ledger is append-only and hash-chained.

Earlier records are not silently rewritten. Forgetting establishes a new active boundary through a nonce-confirmed operation; it does not erase the previous bytes.

A successful turn can produce several different descendants:

1. An exact durable dialogue record.
2. Eligible typed Companion fact, decision, continuity, or open-loop records.
3. A disposable Hyper-RAG projection of verified active records.
4. A staged FRAK conversation candidate.
5. Two durable proposal-learning observations, one for each reaction.
6. A digest-only live-cycle audit record.
7. Read-only membrane or ATL8 observer evidence.

These descendants do not all become dynamic assembly state.

In particular:

- Hyper-RAG is rebuildable and non-authoritative.
- FRAK candidates require a separate admission decision.
- Companion recall can condition proposals but cannot directly mutate StepCore.
- proposal-learning remains proposal-only.
- Telemetry and audit records cannot re-enter the current transition.
- Final answer text is not semantic truth merely because it was stored.

The process-local dialogue window disappears when the shell exits. Durable Companion records do not.

1.7 What UMBRA Does While Nobody Is Talking

Continued inactivity moves the environment through bounded phases.

During ordinary idle operation, the dynamic assembly continues to tick.

At shorter-idle entry, UMBRA can:

- verify the Companion chain;
- append compression or supersession records for eligible memory;

- reconcile rebuildable Companion and FRAK indexes;
- verify the immutable AEML chain;
- and verify the current hot SEMU view.

During extended-idle processing, it additionally:

- verifies retained SEMU dynamic assemblies;
- widens the proposal-only FRAK geometric query window;
- samples finalized post-StepCore geometry;
- and records bounded match or no-match idle maintenance events.

T16CRE continues moving the Polylith during these phases.

No idle-maintenance language-model call occurs. The idle-maintenance pass cannot rewrite AEML, silently alter prior memory, admit a FRAK proposal, or bypass StepCore.

A new geometric association discovered during the idle-maintenance pass may be offered once, through the bounded later recall context of a successful conversation. It remains an association candidate.

This is implemented idle computation called an idle-maintenance pass. It is not evidence of a subjective mode.

1.8 What the Visible Presentation Artifact Represents

ATL8 receives finalized, authenticated dynamic assembly evidence and renders it.

The present paired launcher selects the experimental White Coma profile: a horizonless muted-white field containing a dark onyx Polylith dynamic assembly.

The dynamic assembly surface is derived from the current PSM-8 field and morphotype state. Internal colour and luminosity are derived from the sixteen T16CRE concentration channels and their spatial Laplace-Beltrami structure. Exact zero chemistry returns literal dark onyx.

Prepared morphic surfaces can interpolate between authenticated endpoint fields. Some intermediate fields may undergo topology changes. The 60 Hz presentation thread traverses prepared surfaces; it does not perform the canonical dynamic assembly transition.

ATL8 can display:

- the current form;
- internal chemical emission;

- selected reaction activity;
- dynamic assembly tick;
- kinetic and morphological evidence;
- and read-only diagnostic information.

It cannot:

- choose a morphotype;
- alter a StepCore input;
- change a reaction;
- admit CSL;
- write memory;
- govern the camera back into the dynamic assembly;
- or turn visual interpretation into canonical state.

The visible presentation artifact is therefore a presentation of the simulated dynamic assembly, not a second dynamic assembly and not its sovereign controller.

The presentation remains experimental. Current refinement does not possess a final post-change Forge/video acceptance witness.

There is also a known morphology-selection bias in preserved live evidence: two morphotypes dominated one measured run, while the radial-work selector remained negative throughout that run. The renderer displays the committed result; it does not correct or conceal this bias.

The desired future elastic coupling from authenticated T16CRE torque and impulse through RT32 Hodge observables into distributed membrane deformation is not implemented. The current dynamic assembly may consequently appear materially stiffer and less presentation artifact-like than the intended future system.

1.9 Media, Speech, and Source Inspection

UMBRA can accept explicit local media commands:

```
/ingest-book PATH
/ingest-image PATH
/ingest-video PATH
/ingest-audio PATH
```

The source is read incrementally through the shared Tokamak. Raw bytes are content-addressed and verified before a completion receipt is issued.

Depending on the medium, UMBRA can perform bounded:

- UTF-8 or PDF text extraction;
- image or video frame decoding;
- audio decoding;
- rolling frame or sample hashing;
- recorded-speech recognition;
- and model-assisted semantic observation.

Media intake yields to operator input and active dynamic assembly transactions. It pauses during the conversational model/dynamic assembly cycle.

Completed media reports can condition the final response only after Reaction B. They do not enter Reaction A or Reaction B, and they do not automatically become Companion, FRAK, SEMU, or AEML memory.

A staged media-derived FRAK candidate requires explicit inspection and admission.

Processing a video does not prove that UMBRA watched it. Decoding audio does not prove that UMBRA heard it. Producing a semantic report does not prove understanding.

The current continuation also provides a repository-bundled read-only source MCP. The operator can ask UMBRA to inspect a named component such as T16CRE, ATL8, or the proposal learner. UMBRA receives only a bounded repository-relative excerpt.

The source MCP cannot:

- write files;
- invoke commands;
- run tests;
- commit code;
- admit CSL;
- call StepCore;
- mutate memory;
- or revise UMBRA.

The model may explain what it reads or propose future work. It cannot implement its proposal.

This source-inspection surface is branch-local in the present snapshot and does not yet carry current-head full-suite or hosted publication proof.

1.10 Effective Configuration Truth

Several mechanisms exist in code but are not fully active under the normal live configuration.

The current `membrane_basin_v740.json` specifies:

- 32 index nodes;
- fixed basin and hysteresis thresholds;
- a CSL budget of `0.25`;
- renewal disabled;
- canonical morpho-membrane support disabled.

It does not specify `sigma_min` or `sigma_max`.

The runtime loader therefore uses:

$$
\sigma_{\min}=0
$$

and:

$$
\sigma_{\max}=+\infty
$$

The spectral-corridor regulator is present and traversed by the live StepCore call path, but the default infinite upper bound means it normally supplies no effective upper spectral constraint.

The law is wired. Its upper wall is not currently biting.

Likewise, `renewal_enabled=false` means the canonical renewal mechanism is not active in this live configuration.

The experimental Polylith morphic dynamic assembly still changes through its scoped dynamic assembly machinery. That must not be mislabeled as activation of the disabled canonical morpho-membrane configuration flag.

This is precisely why code existence, runtime traversal, and effective configuration must be reported separately.

1.11 What UMBRA Cannot Honestly Claim

At this snapshot, UMBRA cannot honestly claim that it:

- is conscious;
- is sentient;
- is biologically alive;
- possesses subjective experience;
- understands every operator sentence;
- independently establishes semantic truth;
- has unrestricted self-directed goals;
- rewrites its own canonical laws;
- autonomously edits its source code;
- gives memory direct authority over StepCore;
- perceives media in the human sense;
- has completed long-horizon adaptive learning;
- possesses a fully converged memory architecture;
- implements the complete historical v7.2 design;
- is independent of an external language-model adapter;
- has solved morphology-selection bias;
- has implemented the intended elastic torque/Hodge dynamic assembly coupling;
- has an active finite upper spectral corridor in the default configuration;
- has active canonical renewal;
- or has been fully validated and published at the current continuation head.

Its language can sound more complete than the machine is.

That is why the evidence, admission, and containment layers exist.

1.12 The Shortest Accurate Portrait

UMBRA is presently a continuously ticking, dimensionless Polylith dynamic assembly whose lawful state is mutated only by StepCore v740.

An operator sentence is deterministically reduced into a bounded symbolic proposal and admitted into that dynamic assembly as Reaction A. Only after Reaction A exists may a local language model inspect the permitted evidence and propose Reaction B. That proposal must cross the same deterministic compiler, firewall, admission boundary, membrane, and StepCore path. Only after Reaction B finalizes may the model answer.

Successful exchanges are sealed into append-only Companion storage. Geometric, associative, vector, idle maintenance, proposal learner, media, and source-inspection components can condition later proposals or explanations, but none receives direct canonical mutation authority.

The presentation artifact shown by ATL8 is a read-only expression of the dynamic assembly's authenticated state. It is neither a decorative model avatar nor a controller.

UMBRA is therefore more than a chatbot wrapped in graphics.

It is also much less than a conscious autonomous system.

What exists now is a governed live prototype situated exactly between those two statements.

---

02. The Executable Substrate

StateV740, StepCore, and Lawful Change

UMBRA's strength does not come from the number of components surrounding it. It comes from the fact that those components cannot silently disagree about who owns lawful change.

At the centre of the system is a small canonical state and one transition authority:

```
StateV740
+ StepInputsV740
+ BasinConfig
→ step_core_v740(...)
→ StepResultV740
```

StepCore does not contain every process occurring inside UMBRA. T16CRE chemistry, RT32 kinetics, Companion memory, Tokamak circulation, Hyper-RAG, the proposal learner, idle maintenance events, and ATL8 presentation all maintain their own bounded data.

They do not constitute alternative canonical identity authorities.

StepCore v740 owns the canonical transition from one `StateV740` to the next. The other components may construct inputs, maintain supporting state, preserve evidence, propose morphology, or observe the result.

This distinction is the foundation of the live system.

2.1 From v7.2 Architecture to Executable Law

Proto-Pineal v7.2 described a broad mathematical architecture organized around:

- the continuous internal field (\Psi(x,t));
- the continuity field (\Phi(x,t));
- a high-dimensional latent state (h_t);
- a dynamic symmetric positive-definite metric (G_t);
- a transition-energy functional (\Delta L);
- a modulation vector (U_\mu);
- identity invariants (N) and (J_\mu);
- CSL symbolic operators;
- Rule Induction Mechanism (RIM) procedural constraints;
- Fractal Field Gradient (FFG) temporal dynamics;
- and a continuous PDE limit.

That architecture is part of Proto-Pineal's intellectual ancestry. It is not reproduced here as if every term were active in UMBRA.

The executable descendant is more specific and more exact.

v7.2 object	**Status in UMBRA v8.1**
Continuous (\Psi(x,t)) field	Historical analytical lineage; not the canonical runtime state
Continuity field (\Phi(x,t))	Not maintained as a separate live canonical field
High-dimensional latent (h_t)	Replaced operationally by the finite `StateV740` substrate
Dynamic metric (G_t)	Replaced by an explicit static metric from `BasinConfig`
Runtime PDE	Not solved and not granted runtime authority
(\Delta L)	Has an implemented StepCore descendant and detailed telemetry
(U_\mu)	No complete live mathematical field; bounded response-allocation receipts recover only a narrow modulation function
(N) and (J_\mu)	Not canonical `StateV740` fields
CSL	Implemented as bounded symbolic proposal input
RIM	Implemented as an external deterministic proposal emitter, not a StepCore regulator
FFG	Implemented as optional non-expansive candidate smoothing

v7.2 object	Status in UMBRA v8.1
Synthetic Continuity Substrate	Preserved as historical architecture and scaffolding; StepCore v740 is the current canonical authority

The historical mathematics remains useful where it explains an implemented descendant or proposes a falsifiable future mechanism. It is not executable merely because the notation exists.

UMBRA is not a literal execution of the v7.2 paper. It is the strongest governed materialization to emerge from that lineage.

2.2 Canonical State

The canonical state at logical tick (t) can be summarized as:

[
X_t =
\left(
t,,
m_t,,
k_t,,
V_t,,
q_t,,
W_t,,
M_t
\right)
]

where:

- (t) is the logical tick;
- (m_t \in \mathbb{R}^{32}) is the index-node-state vector;
- (k_t \in {0,1,2,3,4}) is the current basin;
- (V_t \ge 0) is accumulated vitality;
- (q_t) contains bounded denial, cooldown, and null-step counters;
- (W_t) contains bounded regulatory history windows;
- (M_t) is the optional StepCore-owned Polylith morphology state.

In source, this is `StateV740`.

Index-node state

The primary continuous component is:

$$
m_t =
\begin{bmatrix}
m_t^{(1)} &
m_t^{(2)} &
\cdots &
m_t^{(32)}
\end{bmatrix}^{T}
$$

Each entry is finite and belongs to the configured projection domain.

The index-node array is made non-writeable when the state is constructed. A consumer cannot lawfully mutate the existing array in place and pretend that a StepCore transition occurred.

Basin state

The scalar (k_t) records one of five energy basins.

The live configuration defines the threshold sequence:

$$
\tau =
(0.0,\ 0.40,\ 0.80,\ 1.20,\ 1.80,\ +\infty)
$$

Transitions between basins use separate upward and downward thresholds:

$$
\tau_{\mathrm{up}} =
(0.43,\ 0.83,\ 1.23,\ 1.83)
$$

$$
\tau_{\mathrm{down}} =
(0.37,\ 0.77,\ 1.17,\ 1.77)
$$

The separation creates hysteresis. A state near a boundary does not oscillate between basins merely because its energy fluctuates by a small amount.

For the static metric (G), index-node energy is:

[
E(m_t)=\sqrt{m_t^{T}Gm_t}
]

The current UMBRA configuration does not supply a custom metric diagonal, so:

[
G=I_{32}
]

The implemented geometry is therefore static, symmetric positive-definite, and currently Euclidean over the canonical index-node vector. It is not the dynamically evolving (G_t) proposed in v7.2.

Vitality

`vitality_V` accumulates only on `APPLIED` transitions.

In simplified form:

[
V_{t+1}
=
V_t

- E*{\mathrm{final}}
  \left|
  \Delta m{\mathrm{final}}
  \right|_G
  ]

for an applied transition, while a rejected transition does not increase vitality.

Vitality can widen selected effective tolerances through:

[
s(V)=
1+
V_{\max}
\left(
1-e^{-\kappa V}
\right)
]

with the current configuration using:

$$
[
\kappa=0.05,
\qquad
V_{\max}=0.5
]
$$

This is bounded regulatory history. It is not biological baseline reaction dynamics. T16CRE maintains a separate experimental dynamic surface described later in the paper.

Regulatory windows

`StateV740` also holds bounded windows for:

- identity drift;
- Temporal Modulation Kernel recovery and soft-collapse observations;
- Context Integrity Filter recovery and face concentration;
- Self-Stability Feedback Controller recovery and concentration;
- Projection Face Deconcentration;
- Basin Face Escape;
- renewal denial and cooldown state.

These fields allow enabled regulators to depend on bounded trajectory evidence rather than only the current point.

Their presence in the state does not mean their associated mechanisms are active under every configuration.

Morphology

The optional `polylith_morphology` field is the only Polylith morphology state owned directly by StepCore.

A proposed morphotype transition must bind:

- the expected canonical source-state hash;
- the current morphology;
- the proposed target;
- the morphology catalogue;
- the legal transition graph;
- the proposal digest;
- the adjudication digest;

- and the required physical budget.

A committed morphology change requires an `APPLIED` canonical transition.

The renderer cannot commit one.

2.3 Canonical State Is Not All Runtime Data

UMBRA contains several legitimate forms of state.

State surface	Owner	Role
`StateV740`	StepCore	Canonical transition and identity state
RT32/T16CRE operator-dynamic assembly state	Polylith live environment	Replay-bound experimental kinetics and chemistry
Companion ledger	Companion memory	Durable dialogue and typed response-layer memory
Tokamak chamber and CAS	Tokamak	Bounded transport, pressure relief, and content storage
SEMU/AEML/FRAK structures	Memory Spine	Geometric lineage and proposal-only retrieval
proposal learner ledger	proposal learner	Durable proposal-learning observations
Hyper-RAG index	Hyper-RAG service	Disposable reconstruction of verified memory records
ATL8 snapshot	ATL8 presentation	Latest read-only visible state
Cycle and observer journals	Evidence components	Audit, replay, failure, and falsifiability evidence

Calling StepCore the sole canonical mutation authority does not mean StepCore stores every byte changed by UMBRA.

It means none of these other state surfaces may impersonate a canonical `StateV740` transition.

The T16CRE chemistry can advance. Companion can append a dialogue record. Tokamak can persist a content-addressed cell. ATL8 can replace its latest display snapshot.

None of these operations may directly rewrite `StateV740`.

2.4 Canonical Inputs Are Proposals

StepCore accepts `StepInputsV740`, containing only:

- a drive vector;
- a bounded CSL program;
- a calm-regime flag;
- an authorization result;
- and an optional Polylith morphology proposal.

In mathematical form:

$$
U_t =
\left(
d_t,,
P^{\mathrm{CSL}}_t,,
g_t,,
a_t,,
P^{\mathrm{morph}}_t
\right)
$$

where:

- $d_t \in \mathbb{R}^{32}$ is a proposed index-node drive;
- P^{CSL}_t is an ordered symbolic program;
- g_t requests calm decay when true;
- a_t carries the authority-gate result;
- P^{morph}_t is an optional morphology proposal.

The input object contains no raw operator sentence, model prompt, Companion memory block, Hyper-RAG document, ATL8 image, or final model response.

Those materials must be transformed through their lawful upstream boundaries before any bounded derivative can approach StepCore.

Even then, the input remains a proposal.

StepCore decides whether a candidate is admissible.

2.5 Candidate Construction

Ignoring optional pre-regulators for the moment, the index-node candidate is:

[
\widetilde{m}_{t+1}
=
(1-\lambda)m_t

- \eta_{\mathrm{eff}}d_t
- \Delta_{\mathrm{CSL}}
  ]

where:

- (\lambda) is the decay rate;
- (\eta_{\mathrm{eff}}) is the effective drive scale;
- (d_t) is the admitted membrane-derived drive;
- (\Delta_{\mathrm{CSL}}) is the bounded symbolic contribution.

The default live runtime uses:

[
\lambda=0.05
]

and:

[
\eta=1.0
]

When the calm flag is active, the external drive is omitted:

[
\widetilde{m}_{t+1}
=
(1-\lambda)m_t

- \Delta_{\mathrm{CSL}}
  ]

CSL contribution

The CSL engine executes only an admitted ordered program. It produces a candidate displacement rather than a state write.

The current live configuration enforces:

$$
\left|
\Delta_{\mathrm{CSL}}
\right|_G
\leq 0.25
$$

The engine records:

- executed symbols;
- unscaled displacement norm;
- final displacement norm;
- whether scaling occurred;
- whether the budget was reached;
- and total symbolic budget cost.

A program can therefore execute while having its physical influence reduced to remain inside the symbolic budget.

Execution means that the bounded operator contribution was evaluated. It does not prove that every semantic interpretation associated with the glyph is true.

2.6 Optional Pre-Regulatory Tissue

Before the fixed canonical regulator sequence, StepCore contains several optional mechanisms.

Temporal Modulation Kernel

TMK may reduce the effective step scale:

$$
\eta_{\mathrm{eff}}
=
\alpha_{\mathrm{TMK}}\eta
$$

It observes bounded recovery, gap-concentration, soft-collapse, and cooldown history. It constrains tempo rather than selecting a semantic direction.

Context Integrity Filter

CIF examines chronic recovery-face concentration and related trajectory evidence. Depending on its risk region, it may accept, attenuate, or restrict a candidate.

Its implemented StepCore descendant is a geometric hardening mechanism. This does not exhaust the broader historical meaning of the Cognis 11 Context Integrity Filter.

Self-Stability Feedback Controller

SSFC may leave the candidate normal, clamp its magnitude, or contract it toward the current state under chronic concentration.

It does not commit state by itself.

Fractal Field Gradient

The implemented FFG descendant applies non-expansive Laplacian smoothing to the candidate displacement:

$$
\Delta m' = \operatorname{FFG}_{\gamma}(\Delta m)
$$

It records the displacement norm before and after smoothing and the effective γ.

This is real implemented FFG functionality. It is narrower than the historical multiscale temporal field described in v7.2.

Recovery-only mechanisms

Projection Face Deconcentration and Basin Face Escape participate in recovery handling rather than ordinary semantic steering.

Their role is to prevent a recovery process from repeatedly concentrating on the same projection face or remaining trapped in a basin boundary condition.

All of these mechanisms are subordinate to the canonical regulator sequence and final admissibility decision.

2.7 Fixed Regulator Order

After candidate construction, StepCore applies the canonical regulator order:

```
candidate
→ R_dL
→ R_E
→ R_S
→ R_dSigma
→ sigma-box projection
→ final recomputation
→ shared admissibility
→ APPLIED or NULL_STEP
```

This order is not presentation metadata. Changing it changes the resulting system.

Transition-cost regulation — (R_{\Delta L})

The first regulator evaluates the proposed displacement using the configured metric, current basin, barrier structure, and potential basin transition.

It records components including:

- kinetic displacement cost;
- basin barrier contribution;
- basin-jump contribution;
- total raw transition cost;
- total final transition cost;
- and whether scaling occurred.

When a finite (L_{\max}) is configured, the regulator contracts the candidate until the transition cost becomes admissible or its bounded correction process is exhausted.

It cannot enlarge the candidate.

Energy regulation — (R_E)

The energy regulator evaluates the post-(\Delta L) displacement.

With a finite (E_{\max}), it scales excessive deformation and records:

- raw energy;
- final energy;
- scale factor (\alpha_E);

- overflow heat;
- whether scaling occurred;
- and whether mutation remains permitted.

It operates on the result of (R_{\Delta L}), not on the original candidate.

Identity-strain regulation — (R_S)

The strain regulator receives the post-energy displacement.

With a finite effective (S_{\max}), it contracts identity strain and records:

- raw strain;
- final strain;
- scale factor (\alpha_S);
- and whether scaling occurred.

Low transition energy does not automatically imply low identity strain. The measurements remain distinct.

Spectral regulation — (R_{\Delta\Sigma})

The spectral regulator evaluates the displacement against a configured Laplacian and spectral corridor:

[
\Sigma*{\min}
\leq
\Sigma
\leq
\Sigma*{\max}
]

If the candidate lies outside the corridor, the regulator performs a bounded contraction process. If it cannot reach the permitted region within its iteration budget, it freezes the spectral contribution and records the failure.

Telemetry retains:

- raw and final spectral values;
- lower and upper violations;
- contraction factor;
- iteration count;

- scaling status;
- and spectral-freeze status.

Absorbing box projection

The regulated candidate is finally projected into the configured index-node domain:

[
m*{t+1}^{(i)}*
=
min
left(
b_i,,
max(-b_i,m{\mathrm{candidate}}^{(i)})
\right)
]

The normal UMBRA configuration resolves (b_i=1) for all 32 index nodes.

Projection is deterministic and idempotent. StepCore records whether clipping occurred and how many index-node entries were clipped.

Projection is not treated as automatic legality. All relevant final quantities are recomputed after projection.

2.8 Final Admissibility

StepCore does not assume that a regulated and projected candidate is lawful.

It recomputes:

- index-node energy;
- destination basin;
- transition cost;
- energy and strain;
- spectral position;
- identity displacement;
- identity-window state;
- hard and soft collapse indicators;
- authorization;
- CSL budget compliance;

- and morphology admissibility.

The shared admissibility decision can reject a candidate for reasons including:

- unauthorized transition;
- exhausted transition-cost correction;
- energy failure;
- strain failure;
- spectral failure;
- post-projection illegality;
- symbolic budget failure;
- hard identity collapse;
- persistent soft-collapse evidence;
- or illegal morphology transition.

No downstream model or observer may override this decision.

2.9 `APPLIED` and `NULL_STEP`

StepCore exposes two canonical transition outcomes.

`APPLIED`

An `APPLIED` result means that at least one lawful candidate was accepted and a canonical component changed.

The result reports:

```
committed=True
rollback_used=False
```

The index-node vector, morphology, or both may change. Vitality may increase. The new state becomes the sole canonical successor.

`NULL_STEP`

A `NULL_STEP` means that candidates were considered but none were admissible.

The result reports:

```
committed=False
rollback_used=True
```

The rejected index-node deformation and morphology proposal do not commit.

A null step is not the absence of time. The logical tick, null-step streak, bounded regulatory windows, and lineage hash may still advance. What remains unchanged is the rejected physical index-node or morphology candidate.

This distinction preserves evidence that an attempted transition occurred without pretending that the attempted deformation became lawful state.

Recovery

After a configured null-step streak, the core may construct a bounded recovery candidate. Recovery remains subject to the same final legality conditions.

A recovery attempt may therefore:

- commit as an `APPLIED` transition;
- fail and remain a `NULL_STEP`;
- or produce explicit recovery diagnostics without changing the dynamic assembly.

Recovery is not an emergency bypass.

2.10 Hash Identity and Replay

Every `StateV740` field participates in the state hash.

The state is serialized into canonical JSON with:

- sorted keys;
- stable separators;
- explicit schema identity;
- finite numeric values;
- ordered state windows;
- all index-node entries;
- all counters;
- vitality;
- basin;
- logical tick;

- and the complete committed morphology payload when present.

The resulting SHA-256 digest is:

[
H_t
=
\operatorname{SHA256}
\left(
\operatorname{CanonicalState}(X_t)
\right)
]

Every finalized `TelemetryV740` record binds:

[
H*t*
rightarrow
H{t+1}
]

through `prev_state_hash` and `post_state_hash`.

Telemetry also records the final committed index-node displacement:

[
\Delta\psi*{32}*
=
m{t+1}-m_t
]

after regulation and projection.

This supports deterministic replay when the following are preserved:

- exact pre-state;
- exact configuration;
- exact admitted input;
- exact CSL program;
- exact morphology proposal;
- explicit runtime parameters;
- and compatible numerical execution.

The external language model is not inside this deterministic guarantee. Its generated proposal can vary. Once a particular proposal has been deterministically compiled and admitted as an exact StepCore input, the canonical transition no longer depends on the model’s hidden internal state.

Replay is strongly tested within the governed environment. It is not presented as a proof of bit-identical numerical behavior across every possible processor, BLAS implementation, NumPy version, or future runtime.

2.11 Effective UMBRA Configuration: Wired Regulators, Inactive Upper Walls

The ordinary UMBRA launcher uses the following canonical defaults.

Mechanism	**Effective value**
Index-node count	32
Metric	(I_{32})
Index-node bounds	([-1,1])
Decay (\lambda)	0.05
Step scale (\eta)	1.0
CSL displacement budget	0.25
Basin hysteresis	Active
Vitality coupling	Active
(L_{\max})	(+\infty)
(E_{\max})	(+\infty)
(S_{\max})	(+\infty)
(\Sigma_{\min})	0
(\Sigma_{\max})	(+\infty)
FFG	Disabled in this configuration
TMK	Disabled in this configuration
CIF StepCore descendant	Disabled in this configuration
SSFC	Disabled in this configuration
PFD	Disabled in this configuration
BFE	Disabled in this configuration

Mechanism	Effective value
Renewal enforcement	Disabled
Canonical morpho membrane	Disabled
PSM-8 morphology proposal path	Active through separate StepCore adjudication

These values require careful interpretation.

> **Publication-critical distinction:** The (R*{\Delta L}), (R_E), (R_S), and (R*{\Delta\Sigma}) regulatory corridors are fully wired into the selected StepCore code path. In the ordinary baseline run, their upper parameters are (+\infty). They therefore execute and measure, but they do not act as finite upper boundary walls.

Question for the baseline run	Answer
Does the regulator implementation exist?	Yes
Does the live StepCore path traverse it?	Yes
Does it compute and retain measurements?	Yes
Does its configured upper ceiling contract a candidate?	No, because the upper ceiling is (+\infty)

In particular, (\sigma_{\max}=+\infty) means that the spectral corridor has no finite upper spectral wall in this configuration. The same interpretation applies to the infinite upper path-length, energy, and strain ceilings. A future run with finite, explicitly recorded values would be a different effective configuration and would require its own evidence.

Their telemetry remains meaningful as measurement. Their configured upper walls are not presently active constraints.

The live core still actively enforces:

- input and state validation;
- authority decisions;
- the finite CSL budget;
- basin hysteresis;
- finite index-node values;
- the absorbing index-node box;
- identity-collapse predicates;
- morphology adjudication;
- exact transition classification;

- state hashing;
- and replay lineage.

The disabled mechanisms are neither fictional nor secretly active. They are implemented regulatory configurations that are not selected by the ordinary UMBRA basin file.

A paper describing their algorithms may call them implemented. It may not claim that they constrained a particular live tick without corresponding effective configuration and activation telemetry.

2.12 Constitutional Meaning

The executable substrate establishes six laws for the rest of UMBRA.

Law 1 — Proposal is not mutation

A language model, symbolic compiler, memory retriever, proposal learner, or geometry selection controller may construct a proposal.

Only StepCore can turn an admissible proposal into canonical successor state.

Law 2 — Telemetry is not state authority

`TelemetryV740` reports the completed transition. ATL8, memory systems, response policy, and falsifiability instruments may inspect it.

Observation does not grant permission to rewrite its source.

Law 3 — Regulator order is identity-bearing

The sequence of regulators affects the result. Reordering them is not an optimization; it is an architectural change requiring separate authorization and replay consequences.

Law 4 — Rejection remains visible

An inadmissible candidate does not disappear from history. It produces a `NULL_STEP` with explicit reasons, hashes, counts, and regulator evidence.

Law 5 — Configuration is part of truth

Implemented machinery, traversed machinery, and effectively constraining machinery are three different claims.

A regulator with an infinite ceiling exists and executes, but it does not impose a finite upper wall.

Law 6 — Supporting state remains subordinate

T16CRE, memory, Tokamak, proposal learner, idle maintenance mode, Hyper-RAG, source inspection, and ATL8 may evolve within their jurisdictions.

None may present its own evolution as a canonical `StateV740` transition.

2.13 The Actual Foundation of UMBRA

The executable core is more specific than the total conceptual mathematics accumulated across Proto-Pineal's history.

That narrowing is not a reduction in the importance of UMBRA.

It is what makes UMBRA possible.

The live prototype can combine a perpetual dynamic assembly, reaction chemistry, morphology, symbolic language, memory, idle maintenance activity, learned proposals, local language models, media processing, source inspection, and visual expression because the architecture does not allow every intelligent-looking component to become sovereign.

The governing relation is:

$$
\boxed{
\text{Many components may influence proposals;}
\quad
\text{one component commits canonical state.}
}
$$

That component is StepCore v740.

Everything that follows in this paper—the Polylith dynamic assembly, CSL, T16CRE, memory, the proposal learner, idle maintenance activity, ATL8, and the two-reaction conversational cycle—must be understood relative to this boundary.

Executable Anchors

```
src/proto_pineal/core/state_v740.py
src/proto_pineal/core/step_core_v740.py
```

```
src/proto_pineal/runtime/tick_v740.py
src/proto_pineal/runtime/unified_runtime_v740.py
src/proto_pineal/runtime/polylith_live_environment_v01.py
configs/membrane_basin_v740.json
```

---

03. The Polylith Dynamic Assembly

3.1 A Dynamic Assembly Made of Explicit Relations

The Polylith is UMBRA's persistent simulated dynamic assembly. It is not the language model's hidden state and it is not an animation authored after an answer. It is a bounded runtime composition whose state advances on its own clock and whose lawful canonical effects must still traverse StepCore.

Its geometric scaffold is RT32: a fixed rhombic-triacontahedral topology with 32 vertices, 60 edges, and 30 faces. The topology supplies stable addresses for index-node state, differential measurements, force, flux, chemistry, morphology, and evidence. It does not claim that the geometry is biologically privileged or physically fundamental.

The 32 vertices are organized as sixteen antipodal outer/inner pairs:

[
(O*1,I_1),\ldots,(O*{16},I_{16})
]

This paired structure lets the runtime distinguish common-mode and differential activity. The fixed topology also permits deterministic edge orientation, divergence, circulation, Laplace-Beltrami, Hodge-related, and strain witnesses without changing node identity from tick to tick.

RT32 is therefore both a dynamic assembly scaffold and an evidentiary coordinate system.

3.2 Operator Dynamic Assembly and Channels

The experimental operator dynamic assembly maintains a replay-bound state over all 32 RT32 nodes. Each node has a bounded channel coupled to the sixteen-channel chemical system.

An admitted CSL program does not directly choose a visible pose. Its glyph-law plan materializes bounded RT32 force, oriented edge flux, and T16 element bias. These signals enter the operator dynamic assembly and chemistry, which then evolve according to their own finite rules.

The dynamic assembly retains the distinction among:

- a newly secreted element;
- continuing residual chemistry;
- current force and displacement;
- solvent transport;
- reaction production;
- and the finalized geometry derived from those processes.

This distinction matters during silence. No new glyph needs to execute for previously secreted material, velocity, strain, or recovery to continue changing.

3.3 Turing-16 Chemical Reaction Engine

The Turing-16 Chemical Reaction Engine (T16CRE) carries sixteen named dimensionless elements:

`Diluent`, `Coagulant`, `Diffuser`, `Concentrator`, `Vector`, `Vortex`, `Burst`, `Crush`, `Arc`, `Ground`, `Shear`, `Bind`, `Sever`, `Catalyst`, `Inhibitor`, and `Precipitate`.

Each element has an explicit half-life, mobility, and diffusion coefficient. The state contains bounded concentrations, particle records, emission and reaction events, ancestry, solvent displacement and velocity, edge transport, and replay identity.

The current engine adds typed physical profiles to its reaction laws. A reaction can produce bounded radial impulse, tangential impulse, torque, pressure, charge, and edge momentum. Particle records age, move, react, and retire under explicit limits. Small unreacted remainders remain in solution until they are lawfully aged out; they are not silently deleted merely because they fall beneath a reaction floor.

T16CRE also maintains a compact dynamic state with bounded excitation, fatigue, recovery, phase, and basal drive. These variables alter the experimental dynamic assembly's continuing activity and are hash-bound into replay evidence.

The word *chemical* names the engine's reaction-and-transport organization. The values are not laboratory concentrations, the reactions are not established chemistry, and the dynamic variables are not measurements of a physical system.

3.4 PSM-8 Geometry selection

PSM-8 is the active experimental Polylith geometry selection controller. It operates over a catalogue of 338 morphotypes connected by a one-edit graph. Its genesis morphotype is `PM0R049`.

The current committed morphotype supplies structural inertia. Exact T16CRE chemistry and bounded RT32 displacement, velocity, force, pressure, strain, and differential signals contribute to an always-on morphogenic field. That field can change which already-lawful outgoing graph edge PSM-8 proposes.

The proposal remains inert until StepCore adjudicates it. PSM-8 cannot commit a morphology change by itself.

The active field-owned expression evaluates a bounded volumetric field and extracts a zero surface through a pinned marching-cubes procedure. The committed catalogue shell remains separately replayable as morphology evidence. Extracted positions and triangles are disposable expression endpoints: they can change every tick and may cross topology without becoming a second canonical morphology state.

This distinction prevents three objects from being confused:

1. the canonical StepCore morphology decision;
2. the continuously expressed Polylith surface;
3. the ATL8 rendering of that surface.

The first is canonical. The second is experimental dynamic assembly expression. The third is read-only presentation.

3.5 Continuous Live Kinetics

The ordinary launcher requests one dynamic assembly tick every 0.25 seconds. A background dynamic assembly thread continues while the terminal waits for input or a model call completes.

Each finalized `PolylithLiveFrameV01` binds the important layers of one dynamic assembly advance: canonical transition evidence, CSL receipts, RT32 state, T16CRE state, differential readings, morphology, expressed geometry, and replay digests.

The dynamic assembly’s continuity does not depend on every tick changing every surface. A lawful tick may preserve morphology while chemistry moves, preserve index-node state while residual chemistry decays, or yield a StepCore `NULL_STEP` while logical time and supporting evidence advance.

The runtime also contains pressure-divertor and asynchronous evidence mechanisms. Audit-critical and scientific evidence fail closed under their defined saturation or writer-error conditions; disposable presentation snapshots may be dropped without killing the dynamic assembly. This distinction was strengthened after live snapshot loss and maintenance failures proved that an observer-side filesystem error could otherwise terminate the continuing continuous dynamic loop.

3.6 What the Dynamic Assembly Does Not Yet Do

The Polylith is the strongest dynamic assembly constructed in the project to date, but several desired physical relations remain absent or incomplete.

- Authenticated T16CRE torque and impulse do not yet traverse a separately established RT32 Hodge law into distributed elastic membrane deformation and recovery.
- The dynamic assembly remains materially stiffer than the intended future presentation artifact.
- Morphotype selection has shown bias toward a small subset of forms in retained evidence.
- The 338-form catalogue is fixed; a broader family of lawful RT topologies would require an explicit correspondence atlas.
- Internal motion does not establish open-world sensorimotor internal dynamic coupling.
- All reported quantities remain dimensionless simulation evidence.

These are limitations of the present dynamic assembly, not evidence that the dynamic assembly is fictitious.

Executable Anchors

```
src/proto_pineal/geom/rt32.py
src/proto_pineal/polylith/rt32_operator_body_v01.py
src/proto_pineal/polylith/t16cre_v01.py
src/proto_pineal/polylith/rt32_stellation_morphogenesis_psm8_v01.py
src/proto_pineal/runtime/polylith_live_environment_v01.py
```

---

04. Cognispheric Symbolic Language and Dynamics-First Transduction

4.1 Language Is Not a Force

UMBRA does not send raw natural language into its dynamic assembly.

The current operator sentence first enters a deterministic natural-language-to-CSL compiler. No model, embedding service, learned classifier, or retrieval component receives that sentence before Reaction A is finalized. This ordering prevents a learned semantic interpretation from secretly becoming the first physical cause.

The compiler normalizes Unicode and tokens, applies conservative inflection and typo handling, matches the longest registered phrases, suppresses negated effects, refuses unsupported conditionals, preserves source order, removes duplicates, and compresses the result to a bounded effect program.

Unknown topic words are retained only through digests. They do not become invented concepts. A sentence may still be classified as a question, correction, disturbance, quiet contact, continuation, or bounded assertion when its subject vocabulary is unknown. `OP_ASSERT` proves only that an utterance occurred; it does not prove understanding or truth.

4.2 The Live Alphabet

The exposed CSL v0.2 alphabet contains 58 identifiers.

Fifty-four identifiers have registered experimental glyph laws. Four are unconditional forbidden sentinels:

- `FORBIDDEN_NEGATION_LOOP`
- `FORBIDDEN_DELTA`
- `FORBIDDEN_XI`
- `FORBIDDEN_OMEGA`

Their presence rejects the whole program. A forbidden identifier cannot be made safe by placing it beside valid operators.

The admitted registry distinguishes direct bounded operators, composed protocols and symbols, and zero-force declarative forms. Zero-force forms remain executable identities and receipts, but they do not fabricate physical stimulation.

The active materialization is explicit. It yields:

$$
[
(f_{32},e_{60},B_{32\times16})
]
$$

where (f*{32}) is RT32 force, (e*{60}) is oriented edge flux, and (B_{32\times16}) is T16 element bias. The result also records per-primitive applicability, raw and applied energy scores, heat, leakage, disturbance, and residual drive.

These are experimental simulation scores, not established physical units.

4.3 Proposal, Firewall, and One-Shot Admission

Successful transduction produces canonical CSL text and an inert `PolylithInertCSLProposalV01`. Object construction has no dynamic assembly effect.

The activation firewall checks the exact ordered program, registry, forbidden identifiers, runtime roles, and bounded applicability. A separate admission operation returns `ADMIT_ONCE`, `HOLD`, or `REJECT`.

Only an exact digest-matching `ADMIT_ONCE` receipt can enqueue the proposal. Reuse, alteration, mismatched tuple and string representations, or a non-admitting disposition cannot execute.

This provides a typed distinction among:

```
recognized language
compiled symbols
firewall-valid program
admitted one-shot proposal
executed bounded effect
committed StepCore transition
```

No earlier line implies a later one.

4.4 Executable Algebra

The live program is an ordered S-expression-derived sequence. Direct operators such as `OP_AMPLIFY`, `OP_DAMP`, `OP_QUERY`, `OP_ROTATE`, and `OP_STABILIZE` map to explicit bounded laws. Protocol and symbolic identifiers expand to registered primitive programs. Structural, agent, and control-flow identities that lack a physical law remain zero-force.

The current branch also permits three closed compile-time macros inside the Common Lisp learning surface: `TWIST`, `PULSE_RELAX`, and `FLATTEN`. They must expand completely into the 54 admitted identifiers. A macro name can never reach StepCore as a new runtime operator.

The CSL displacement reaching canonical candidate construction is bounded by the live budget:

$$
|\Delta_{\mathrm{CSL}}|_G\leq0.25
$$

Parameter-freeze, registry, topology, plan, materialization, proposal, firewall, admission, and execution receipts make the transformation replay-visible.

4.5 Refusal Cases

Transduction or activation fails closed for empty input, punctuation-only input, excessive character or token counts, unsupported conditional structure, material ambiguity, forbidden sentinels, unknown direct CSL identifiers, malformed S-expressions, and admission mismatches.

For an operator-side transduction or admission refusal, the terminal may report:

```
UNRECOGNIZED / NO EVIDENCED EFFECT
```

That statement is exact. It does not say that the input was meaningless. It says the runtime established no admitted dynamic assembly effect.

Executable Anchors

```
src/proto_pineal/service_layer/natural_language_csl_transducer_v01.py
src/proto_pineal/service_layer/natural_language_csl_lexicon_v03.json
src/proto_pineal/csl/glyph_law_registry_v01.py
src/proto_pineal/csl/activation_firewall_v01.py
src/proto_pineal/csl/engine.py
src/proto_pineal/csl/sexpr.py
```

---

05. The Two-Reaction Live Cycle

5.1 One Conversation, Two Dynamic Assembly Contacts

The live cycle is an ordered transaction, not an unconstrained exchange between an operator and a model.

Its principal stage sequence is:

```
operator transduction
→ operator admission
→ Reaction A
→ Proposal learner A
→ Service envelope A
→ Companion context and response policy
→ proposal-model call
→ proposal transduction
→ proposal admission
→ Reaction B
→ Proposal learner B
→ Service envelope B
→ bounded recall
→ final-model call
→ claim containment
→ complete cycle
```

The cycle result retains the exact stage sequence rather than inferring it from timestamps.

5.2 Reaction A

The current operator text is compiled before any learned component sees it. An admitted program crosses the membrane and advances the live Polylith through `UnifiedRuntimeV740.tick_from_service(...)` and StepCore.

Reaction A ends only when a finalized frame exists. The frame becomes the source for deterministic state-derived reduction, dynamic assembly reporting, response-policy measurement, and the first model prompt.

The proposal model receives permitted current text as a labelled secondary semantic context because it must know what it is responding to. That text does not retroactively construct Reaction A.

5.3 Reaction-B Proposal

The first model call has a separate role and a normal maximum of 96 output tokens. It may propose a short Reaction-B description; it may not answer the operator.

The current branch adds a second possible proposal source. When the durable Common Lisp learner has sufficient repeated evidence and emits a firewalled `PROPOSED` hypothesis, that exact bounded program can supply the next Reaction-B candidate through the existing Service path. `INSUFFICIENT`, `DEGRADED`, or `WAIT` leaves the model-proposal path in place.

This does not let Lisp bypass natural-language or symbolic admission. It replaces only the candidate source under explicit conditions. Operator Reaction A, the Service boundary, the firewall, one-shot admission, and StepCore remain unchanged.

5.4 Reaction B and the Right to Answer

The selected candidate is deterministically compiled or validated, admitted, and submitted through the same dynamic assembly path as Reaction A.

Reaction B must finalize before the model receives answer permission. If it fails, the cycle refuses rather than pretending that a complete two-reaction exchange occurred.

A failure after Reaction A does not erase Reaction A. The first dynamic assembly transition remains part of the actual trajectory. The missing Reaction B prevents completion of the conversational transaction, not retroactive existence of its predecessor.

The `LiveBodyABComparisonPacketV01` then records comparable morphology, chemistry, kinetics, and elasticity values. Exact measurement tokens allow narrow dynamic assembly claims to cite the evidence domain that actually supports them.

5.5 Final Context and Containment

The final call’s mandatory core includes the current request, compact A/B evidence, exact CSL and admission receipts, model identity, cycle identity, and response policy. Dialogue, geometric recall, FRAK, Hyper-RAG, media, and source excerpts are optional bounded sections.

When the verified model context is too small, optional sections are removed in a fixed order and the call record lists each omission. The mandatory request and A/B core cannot be discarded. The ordinary launcher currently refuses model configurations above 8,192 context tokens and uses a 512-token final output maximum.

The generated candidate then crosses deterministic speaker-role, unsupported-experience, dynamic assembly-measurement, glyph, renderer-causality, memory, media, repetition, and current-A/B contradiction guards. Supported sentences may survive while unsupported

sentences are excised or narrowly rewritten. One retry is available for a substantive response loop; persistent repetition is contained.

These guards are finite. They reduce known falsehood classes. They do not certify universal truth.

5.6 Cycle Identity

`ProtoPinealLiveCycleResultV01` binds operator-input digest, both transductions, proposals, admissions, frames, proposal learner results, Service envelopes, memory receipts, model-call records, context omissions, final containment, the emitted answer, and the ordered stage sequence into one cycle digest.

Raw operator, proposal, and answer text are not copied into the digest-only cycle audit. Durable conversation text is handled separately by Companion memory after success.

Executable Anchors

```
src/proto_pineal/runtime/proto_pineal_live_cycle_v01.py
src/proto_pineal/runtime/polylith_live_environment_v01.py
src/proto_pineal/runtime/local_model_vessel_invocation_adapter_v01.py
src/proto_pineal/observability/falsifiability_instrument_v01.py
scripts/proto_pineal_live_chat.py
```

---

06. Companion Memory and Continuity

6.1 The Dialogue Ledger

Companion memory is UMBRA’s durable conversational continuity surface. It is append-only and hash-chained. It is not StepCore state and it is not a semantic truth oracle.

After a successful cycle, exact operator and assistant text is encoded as deterministic 4,096-byte chunks. Unique chunks are stored in the Tokamak content-addressed store, while the ledger retains ordered digest recipes, byte counts, roles, turn sequence, provenance, and cycle bindings.

Recall reconstructs exact UTF-8 bytes and verifies store identity, chunk digest, order, total byte count, and whole-text digest. Missing, corrupt, reordered, or foreign-store chunks fail closed. Historical inline dialogue rows remain readable without being rewritten.

The storage identity is deliberately separate from filesystem location. The current launcher retains the historical identity string required by earlier recipes while placing the active run directory and CAS on internal storage. Moving bytes does not silently rewrite identity; changing identity does not silently orphan history.

6.2 Typed Continuity

Companion memory can retain typed facts, decisions, continuity statements, open loops, dialogue boundaries, and a confirmed Companion name. A model may propose a name, but only operator confirmation creates the durable identity record.

The configured Companion voice is currently female and may remain unnamed. Pronouns are resolved relative to the speaking role, not by a global `I` / `you` substitution. This boundary was added after a smaller model exposed that the prior global rule inverted an operator question about the Companion into an answer about the operator.

Typed records are projections of admitted dialogue and explicit operations. They do not convert every stored sentence into belief.

6.3 Recall and Response Policy

The live controller projects active Companion records into micro, meso, macro, and identity horizons. A deterministic response policy selects `PRECISE`, `ASSOCIATIVE`, `CONSOLIDATE`, or `CAUTIOUS` from finalized dynamic assembly measurements, effective configuration, operator request, and typed context.

The policy changes only bounded context allocation. Hard maxima remain finite: twelve semantic entries, four dialogue turns, four FRAK objects, 8 KiB dialogue recall, 4 KiB FRAK recall, and the live-cycle override boundary.

Recall can influence model proposals and answers. It cannot directly create `StepInputsV740`, admit CSL, or mutate the dynamic assembly.

6.4 Correction, Supersession, and Forgetting

The ledger does not edit old records in place.

A correction appends a relation to its target. A supersession creates a new active statement while preserving the prior statement's historical existence. Dialogue projections explicitly exclude correction records even when a correction inherits the target kind; malformed ordinary dialogue still fails closed.

Forgetting likewise creates a new active boundary through a confirmed operation. It does not make earlier bytes cease to have existed.

This is a strong audit property and a limited privacy property. Logical exclusion is not physical erasure.

6.5 What Companion Memory Is Not

Companion memory is not:

- the whole identity of UMBRA;
- an automatically trusted autobiography;
- a canonical dynamic assembly-state database;
- a route around CSL admission;
- a license for the model to claim experiences;
- or proof that remembered text affected a particular transition.

Its lawful role is durable, reconstructable continuity material for later bounded use.

Executable Anchors

```
src/proto_pineal/memory_spine/companion_memory.py
src/proto_pineal/memory_spine/companion_live_adapter_v01.py
src/proto_pineal/runtime/companion_tokamak_live_bridge_v01.py
scripts/proto_pineal_live_chat.py
```

---

07. Memory Spine, Retrieval, and Idle Maintenance

7.1 Several Memory Lineages Exist

Proto-Pineal contains multiple structures bearing the names SEMU, AEML, FRAK, and retrieval. They must not be collapsed into one completed canonical memory architecture.

The active experimental live descendant is `memory_spine/stable_lineage_v741.py`. It samples finalized post-regulator state into bounded stable windows, can compile those windows into SEMU-like retained dynamic assemblies, appends verified headers to an experimental episodic lineage, and exposes proposal-only geometric retrieval. Its source symbols still use `StableEvolutionaryMemoryUnitV741` and `AppendOnlyEvolutionaryMemoryLedgerV741`; these are recorded historical source-name drift. In public prose, SEMU means **Structured Episodic Memory Unit** and AEML means **Append-Only Episodic Memory Log**.

The older top-level `src/semu`, `src/aeml`, and `src/frak` packages remain sidecar or historical lineages. Their repository presence does not make them the active UMBRA route.

Canonical continuity in the broader v740 substrate still belongs to the Memory Spine's episode-ledger, heredity, population, and commit-gate lineage. The live stable-window AEML/FRAK path is additional experimental memory evidence, not a replacement identity authority.

7.2 Compact SEMU and AEML

A stable sample binds finalized state, telemetry, geometry, configuration, and lineage evidence. A stable-window compiler decides whether enough bounded continuity exists to produce a SEMU.

The compact AEML stores permanent hash-chained headers and exact deterministic-gzip content-addressed SEMU dynamic assemblies. Retention is explicit. Reload verifies header lineage and retained dynamic assemblies before exposing the hot view.

SEMU dynamic assemblies can be inspected or projected into noncanonical replay. That replay is read-only and incomplete relative to the full ATL8 canonical-view evidence chain; it cannot install a historical dynamic assembly as current state.

7.3 FRAK

The live FRAK descendant performs bounded geometric and associative retrieval over verified memory candidates. It can join sparse semantic features with exact origin-state postings and stable-dynamic assembly geometry.

Conversation or media-derived association candidates are staged process-locally. Explicit challenge-bound admission is required before an association object is appended. Duplicate admission is a no-op.

Retrieved FRAK material crosses the Tokamak as `MEMORY_RECALL` only after Reaction B and enters final-answer context. It does not influence the current Reaction A, write StepCore, or establish causation merely because two records are geometrically or semantically associated.

7.4 Hyper-RAG

Hyper-RAG projects verified active Companion records into a disposable LanceDB index. Deterministic chunking retains source offsets. Identical text can share one content-addressed vector while preserving occurrence count and representative provenance.

The current stack uses a pinned 384-dimensional multilingual embedding model, bounded dense and lexical retrieval, and an explicit `validate → retrieve → fuse → compose` graph. A changed Companion projection digest forces rebuild; an unchanged chain requires no new embedding.

Recall is capped at six unique chunks and 4 KiB. Operator-role text and active typed semantic records may enter automatic answer context; Companion-role generated text remains available for audit but is excluded from automatic composition after live evidence showed an old generated explanation could echo back as apparent evidence.

The index is rebuildable. The Companion ledger and Tokamak recipes remain authoritative history.

7.5 Idle Maintenance

The current launcher derives idle maintenance and extended-idle thresholds from the dynamic assembly cadence. At the default 0.25-second tick, idle maintenance begins only after approximately twenty minutes of idle time and extended-idle processing after approximately forty minutes. Submitted operator presence resets idle state even when the line produces no dynamic assembly stimulus.

Shorter-idle entry can verify the Companion chain, append eligible compression or supersession records, rebuild disposable indexes, and verify AEML and the hot SEMU view. Extended-idle processing additionally verifies retained SEMU dynamic assemblies and widens the proposal-only FRAK geometric search budget.

T16CRE continues advancing the Polylith throughout these phases. Maintenance, ledger, FRAK, or disposable snapshot failures are contained so they cannot silently kill the dynamic assembly thread.

Idle-maintenance records form a compressed append-only chain containing matches and misses, phase, idle tick, state hash, replay identity, proposal evidence, and authority flags. The newest undelivered association may enter exactly once through the later post-Reaction-B recall boundary of a successful cycle.

No idle-maintenance model call, automatic FRAK admission, AEML rewrite, present-tick feedback, or subjective-idle maintenance claim is authorized.

Executable Anchors

```
src/proto_pineal/memory_spine/stable_lineage_v741.py
src/proto_pineal/memory_spine/frak_associative_memory_v01.py
src/proto_pineal/memory_spine/geometric_recall_v01.py
src/proto_pineal/memory_spine/hyper_rag_v01.py
src/proto_pineal/runtime/polylith_live_environment_v01.py
```

---

08. Rule Induction Mechanism and Bounded Proposal Learning

8.1 From Historical RIM to a Live Bounded Learner

The historical Rule Induction Mechanism was imagined as a broad procedural component capable of discovering, evaluating, and reusing symbolic regularities.

The current UMBRA descendant is narrower. It combines:

- deterministic reduction of finalized dynamic assembly evidence;
- inert CSL candidate construction;
- an append-only learning ledger;
- a bounded Common Lisp learning workspace;
- and, on the present branch, one proposal-only route into Reaction B.

This is a real learned influence path. It is not a second mind, a general Lisp execution service, or an alternative canonical substrate.

8.2 Deterministic Physical Reduction

After each finalized Reaction-A or Reaction-B frame, the proposal learner evaluates local physical channels such as force, residual quantity, edge gradient, divergence, circulation,

Laplace-Beltrami roughness, recovery, and executed glyph receipts.

Each observation carries its own measurement, threshold, measurement law, proposed operator, and unit status. Measurements from unlike channels are not treated as physically commensurate. The current selector compares threshold-relative exceedance and uses the fixed causal priority only to break ties.

An active observation becomes a `LiveSubBrainCSLCandidateV01`. The candidate contains an S-expression record, exact canonical CSL, source measurement, firewall decision, digest, and the explicit truth status:

```
physical_hypothesis_not_semantic_truth
```

The selected `LiveSubBrainResultV01` remains proposal-only. It does not mutate state.

8.3 Durable Learning Ledger

The current continuation adds `sub_brain_learning_ledger.ndjson` beneath the UMBRA run directory.

The ledger is append-only and hash-chained. Finalized A/B programs are classified relative to prior same-channel evidence as `ASSERT`, `SUPPORT`, `CONTRADICT`, or `SUPERSEDE`. Contradicted claims remain in historical bytes but leave the active view. Same-channel replacement supersedes rather than falsely claiming logical contradiction.

Each current record binds the program and its read-only dynamic assembly posture: chemistry hashes, RT32 inner/outer chains, 32-node trace, stellation evidence, and reaction identifiers. The posture enriches provenance. It does not become StepCore identity or ATL8 authority.

Historical rows lacking newer posture fields remain loadable under their historical schema.

8.4 Common Lisp Workspace

The Lisp component consumes a bounded sequence of recent durable learning entries. It starts from an explicit workspace, learns ordered program-and-channel transition counts without look-ahead, and emits validated pattern, prediction, hypothesis, and revision receipts.

It learns on the repeated symbolic program and channel relation rather than treating changing evidence digests as new semantic states. This distinction allowed repeated support to accumulate instead of being destroyed by the uniqueness of every dynamic assembly receipt.

Three closed macro names— `TWIST`, `PULSE_RELAX`, and `FLATTEN` —may appear only inside this compile-time surface. Each must expand into admitted CSL identifiers before leaving Lisp. Control-flow forms and macro identities cannot reach the dynamic assembly as novel runtime glyphs.

The workspace is bounded and reconstructable. It is not an unrestricted REPL exposed to the model or operator.

8.5 Live Reaction-B Influence

Earlier Common Lisp work was replay-only or terminal-shadow evidence. The current branch advances it one step.

When the learner has sufficient evidence and emits a `PROPOSED` hypothesis whose predicted program passes the live Lisp cage and existing activation firewall, that program may become the next Reaction-B candidate. It enters through the existing Service schema and one-shot admission path.

If the learner reports insufficient evidence, degradation, or a wait condition, the ordinary model-authored Reaction-B path remains active.

The live connection is therefore:

```
finalized prior A/B evidence
→ durable proposal-learning ledger
→ bounded Lisp learner
→ firewalled proposed program
→ existing Reaction-B admission
→ existing Polylith and StepCore path
```

There is no direct Lisp-to-StepCore call. Lisp does not alter Reaction A, edit the operator sentence, admit its own proposal, rewrite AEML, move ATL8, or declare its hypothesis true.

8.6 What Learning Means Here

The current proposal learner satisfies a narrow operational definition of learning: retained evidence can change which bounded future proposal is offered.

It does not establish:

- general reasoning;
- autonomous goal formation;
- open-ended code generation;
- universal symbolic induction;
- independent truth adjudication;
- or self-modification of constitutional law.

The learned component is a bounded routine with memory, not a sovereign authority.

Executable Anchors

```
src/proto_pineal/runtime/proto_pineal_live_cycle_v01.py
src/proto_pineal/rim/sub_brain_durable_learning_ledger_v01.py
src/proto_pineal/rim/common_lisp_shadow_v01.py
src/proto_pineal/rim/csl58_rim_v01.py
lisp/umbra-rim/learning-workspace-v01.lisp
scripts/proto_pineal_live_chat.py
```

---

09. Unified Tokamak and Bounded Ingress

9.1 One Shared Circulation Chamber

The Unified Tokamak is UMBRA's bounded transport and retention chamber. It is not a physics simulation of a fusion reactor and it is not a canonical state authority.

The ordinary live composition constructs one `UnifiedTokamakCirculationV01` and injects it into the dynamic assembly, Companion bridge, media adapters, recall paths, response policy, and model-context transport. When this chamber is injected, the Polylith environment does not construct a private alternative.

This single-chamber rule makes cross-component transport inspectable. It does not mean every payload becomes mutually accessible or semantically equivalent.

9.2 Typed Lanes and Admission

Tokamak ingress uses typed envelopes carrying data class, authority class, retention class, privacy class, payload digest, source evidence, and lane identity.

The chamber applies finite lane budgets and capacity policy. Admission can accept or hold a payload. Phase-change and divertor operations move or release retained material through explicit receipts. Checkpoints bind resident cells, payload slabs, counters, lane occupancy, policies, and digests.

Payload type remains load-bearing. `MEMORY_RECALL`, `SERVICE_ENVELOPE`, dynamic assembly telemetry, media chunks, dialogue storage, and shutdown state do not gain each other's permissions merely by sharing the chamber.

The Tokamak transports bytes and receipts. It does not validate semantic truth, admit CSL, or invoke StepCore.

9.3 Content-Addressed Storage

The Tokamak content-addressed store holds verified dialogue chunks and ingested objects. Objects are split into deterministic chunks with object and chunk digests. Reconstruction validates ordering, expected size, and full digest.

Store identity and store location are separate. The current launcher keeps a stable historical store identity while locating active data on internal NVMe. The identity is checked when dialogue recipes are read; a mismatch fails closed instead of silently substituting another byte store.

This is provenance discipline, not a claim that a UUID string possesses identity by itself.

9.4 Seekable Media Ingress

The commands `/ingest-book`, `/ingest-image`, `/ingest-video`, and `/ingest-audio` create bounded local-source tasks.

The seekable ingress adapter resolves a regular local file, fingerprints its metadata, reads incrementally, and verifies that the source did not change during ingestion. Accepted chunks cross the shared chamber. Cancellation leaves the source untouched and preserves already accepted Tokamak lineage.

The decode adapter can perform bounded text or PDF extraction, frame decoding, audio decoding, and rolling hash observation. The semantic adapter can produce model-assisted text, visual, or speech observations under explicit budgets. Recorded-speech recognition uses pinned local assets when configured.

Operator input and an active dynamic assembly transaction take priority. Media workers pause rather than competing without limit with the conversational cycle.

Completed reports can enter final-answer context only after Reaction B. They do not construct the current dynamic assembly reactions and do not automatically become Companion, SEMU, AEML, or FRAK memory.

Processing evidence proves decoding or analysis operations. It does not prove human-like seeing, hearing, reading, or understanding.

9.5 Read-Only Source Inspection

The present continuation supervises one repository-bundled MCP child over newline-delimited standard input/output. It exposes exactly one tool:

```
umbra.read_source_excerpt
```

Ordinary operator language can name a catalogued component, such as T16CRE or the proposal learner, and stage at most 80 lines for the current final-answer cycle. `/self-review PATH START END` remains the surgical operator-controlled route, within a larger fixed maximum.

The reader is confined to the repository root, validates paths and bounds, records Git and excerpt identity, and returns a digest-bound `ReadOnlySourceReviewContextV01`.

It cannot write files, invoke commands, run tests, acquire Git write locks, commit, mutate memory, admit CSL, call StepCore, or implement a proposal. It creates explanation and proposal material only.

Extended-idle self-inspection is not implemented.

This MCP surface is branch-local at the manuscript snapshot. Focused validation exists; current-head full-suite, push, merge, and hosted CI evidence do not.

9.6 Failure Containment

Tokamak and ingress failures are explicit rather than silently dropped. Invalid envelopes, capacity violations, changed sources, malformed decoder output, failed child requests, and digest mismatches produce typed held or failed receipts.

Failure after a completed conversational cycle does not authorize retroactive mutation of that cycle. Disposable media or source context can be omitted. Critical dialogue reconstruction fails closed. Independent shutdown finalizers are each attempted even when an earlier finalizer fails, with the first failure re-raised only after the remaining sealing operations receive an attempt.

Executable Anchors

```
src/proto_pineal/runtime/unified_tokamak_circulation_v01.py
src/proto_pineal/runtime/unified_tokamak_divertor_v01.py
src/proto_pineal/runtime/seekable_bulk_tokamak_ingress_adapter_v01.py
src/proto_pineal/runtime/bounded_media_decode_adapter_v01.py
src/proto_pineal/runtime/bounded_media_semantic_adapter_v01.py
src/proto_pineal/runtime/umbra_source_review_mcp_v01.py
```

---

10. ATL8, Expression, and Falsifiability

10.1 Expression Without Re-Entry

ATL8 is UMBRA's read-only presentation and inspection surface.

It receives finalized, authenticated evidence. It may interpolate presentation between exact endpoints, select visual channels, display diagnostics, and preserve observer records. It may not select a dynamic assembly transition, alter StepCore input, admit CSL, write memory, or send camera and rendering state back into the Polylith.

The governing direction is:

$$
[
\text{finalized dynamic assembly evidence}
\rightarrow
\text{ATL8 presentation}
]
$$

There is no lawful reverse arrow.

10.2 The Onyx Mantle

The current visible presentation artifact is the Onyx Mantle: one permanent subdivision-four display manifold with 2,562 vertices and 5,120 triangles.

It is not a literal display of one of the 338 PSM-8 catalogue meshes. Those meshes and the field-owned zero surface remain hidden dynamic assembly and analysis evidence. Their current geometry, differential fields, chemistry, impulse, and torque are projected into bounded pressure sites and local deformation over the continuous mantle.

The presentation manifold uses under-damped vertex motion, a rest spring, softened neighbour coupling, and velocity viscosity. Up to 40 ordinary or 72 changed-form pressure contacts can produce local tenting, travelling ridges, ripple, local bulge, twist, and wobble. When pressure withdraws, the same continuous membrane law owns recovery.

The current continuation explicitly removes event-synchronized global expansion, compression, stellation tension, and inward release kicks. Those mechanisms made the whole dynamic assembly pulse on reaction cadence and were rejected as presentation pretending to be dynamic mechanics.

The intended future torque/Hodge/elastic dynamic assembly law remains absent. Present local expression is ATL8-side pronunciation of authenticated evidence, not a new physical state.

10.3 White Coma

The ordinary paired launcher selects the `white-coma` profile.

White Coma is a static horizonless near-white field with dark graphite instrumentation and a black wet-mirror presentation artifact. One wider neutral studio reflection and a fixed presentation-only lumen lift reveal internal T16 colour through the onyx material.

Exact zero chemistry produces literal dark onyx. Nonzero chemistry uses all sixteen authenticated concentration channels and spatial Laplace-Beltrami structure. The optical mapping is bounded and cannot change T16CRE identity or magnitude.

The room, reflection, light, camera, and colour transfer are presentation policy. Telemetry cannot move the studio light.

10.4 Presentation Motion

ATL8 also applies a deterministic bounded camera/dynamic assembly presentation trajectory through lateral, vertical, and depth axes. It can retreat, traverse distant space, approach, pause, and return. Authenticated reaction torque can add bounded orientation influence.

This motion is deliberately expression-only. It is not Polylith locomotion, world navigation, independent desire, or a dynamic assembly-state transition. The distinction must remain visible because an expressive camera can easily be mistaken for agency.

The 60 Hz UI samples prepared presentation state. It does not advance StepCore, T16CRE, or the live dynamic assembly clock.

10.5 Observer and Failure Evidence

The launcher starts a separate read-only membrane observer. ATL8 maintains timing summaries and a parent-owned append-only callback-failure journal. The parent records newly observed worker failures so a dead hydration worker cannot erase the evidence of its own failure.

Live recordings exposed several real defects:

- a roughly 155-second presentation freeze while dynamic assembly ticks and conversations continued;
- strict rejection of a one-ULP internal-emission overflow;
- repeated indexed-face colour corruption;
- interaction between blended reflection lines and live signals;
- large UI stalls from hydration and topology work;
- a visible ring that read as an overlay rather than dynamic assembly structure;
- synchronized global pulses that obscured local dynamic assembly expression.

These were not reclassified as aesthetic preferences after the fact. They remain negative evidence explaining why the current surface differs from earlier versions.

10.6 Falsifiability Instrument

The falsifiability instrument classifies known claim types against the evidence available to the completed cycle.

Its governing law is:

> A claim is scientific if you can name the miss.

It checks recognized unsupported experience, dynamic assembly, glyph, media, memory, speaker-role, renderer, and current-cycle claims. Each finding records whether the claim passed or was caught, the relevant evidence sheet, and a digest.

The instrument treats the mouth as suspect because fluent final language can exceed the system that produced it. It does not claim that every possible lie has been enumerated.

ATL8 and falsifiability therefore serve complementary roles: one makes authenticated state visible; the other records when language outruns authenticated evidence. Neither becomes causal authority.

Executable Anchors

```
src/proto_pineal/observability/atl8_live_causal_field_v01.py
tools/atl8_live_causal_field_v01.py
src/proto_pineal/observability/umbra_live_membrane_observer_v01.py
src/proto_pineal/observability/falsifiability_instrument_v01.py
scripts/talk_to_umbra.sh
```

---

11. Validation, Failure, and Negative Results

11.1 Evidence Is Layered

UMBRA's validation record is not one number.

The relevant layers are:

1. source and schema existence;
2. focused tests of a named property;
3. honest-red demonstration that the guard detects its violation;
4. wider affected test matrices;
5. a complete suite on an exact clean commit;
6. hosted checks on that exact commit;
7. merge and ancestry proof;
8. a real live witness under the selected launcher and hardware;
9. retained evidence proving the claimed behavior occurred.

Success at one layer does not imply success at the next.

11.2 Published Baselines

Several major ancestors have exact merged proof:

Surface	Publication boundary	Reported hosted test result
RT32/T16CRE full-cycle dynamic assembly	PR #290	9,606 passed, 2 skipped
ATL8 live causal field	PR #293	9,670 passed, 2 skipped
T16CRE kinetics and presentation	PR #294	9,706 passed, 2 skipped
UMBRA cognition and morphic-dynamic assembly stack	PR #298	10,296 passed, 2 skipped
Middle-out continuation	PR #300	10,394 passed, 2 skipped

These counts are historical artifact evidence. They are not additive, and they do not validate code added later.

The record also preserves honest hosted failures. PR #298's first hosted head failed because a direct-chat Hyper-RAG default named a developer-local home path. The repair made the default portable while retaining explicit Forge paths. PR #300's first hosted head failed because a subprocess test inherited developer LM Studio discovery. The repair supplied a deterministic loopback fixture without weakening production startup refusal.

The failures matter because the later green runs are intelligible only in relation to what first broke.

11.3 Present Snapshot Boundary

The manuscript snapshot is commit:

```
687406001db896413003b614720067b0b9bf4f82
```

Its tree is:

```
b3540eaa1477221d446d98d8ea94dcf18c964568
```

This continuation contains later restart, idle-maintenance, durable proposal-learning, Common Lisp, presentation, source-MCP, context, and automatic-model-selection work beyond the merged PR #300 baseline.

Focused evidence includes, among other non-additive matrices:

- 305 proposal learner family tests after posture binding;
- 185 model-context and live-terminal tests for the 8,192-token startup boundary;
- 239 ATL8/live-environment/PSM-8/doctrine tests for the White Coma and mantle refinement;
- 177 ATL8 tests after removal of the remaining synchronized global pulse;
- 172 MCP/live-cycle/terminal tests for ordinary-language component inspection.

Earlier in this continuation, exact clean commit `09b8fe9` completed the full repository suite with 10,413 passed and 2 skipped after a falsifiability classification repair.

That result does not cover the later commits through `6874060`.

At the manuscript snapshot there is no claimed exact-head full-suite result, hosted CI result, push, pull request, merge into `main`, or post-refinement live acceptance witness.

11.4 Honest Red

A severe guard counts as enforcement only when breaking its claimed property makes it fail.

The UMBRA series includes honest-red demonstrations for one-shot proposal reuse, forbidden glyph admission, context omission, storage identity, correction projection, restart cycle identity, dead dynamic assembly reporting, idle-maintenance containment, proposal learner supersession, Lisp macro closure, Lisp live admission, response-loop containment, source-reader confinement, model auto-selection, and ATL8 local-expression laws.

It also contains examples of phantom guards. One ATL8 test passed while the diagnostic cause was lost because it exercised a path that already drained the worker queue. Another older assertion required an unwanted global tension scale. Both cases show why a green test name is not enough: the mutation must falsify the law the test claims to bind.

11.5 Live Evidence and Its Limits

Real launcher, LM Studio, KDE/X11, VisPy, replay, media, and ledger witnesses exist across the lineage. They demonstrate such properties as dynamic assembly continuation during model calls, exact A/B order, renderer liveness after repair, bounded media ingestion, dialogue reconstruction, and visible presentation.

No single witness proves the whole system. A video can show visible behavior without proving internal authority. A ledger can prove order without proving subjective interpretation. A deterministic fake model can prove the pipeline without proving the behavior of an arbitrary local model.

Live evidence is strongest when the claim names the exact run, configuration, model, frame or cycle digests, and retained artifacts.

11.6 Known Negative Results

The current design was shaped by failed or rejected work:

- model-driven semantic analysis before Reaction A violated dynamics-first causality and was removed from the live path;
- schema examples caused an earlier learned analyzer to echo a tiny glyph vocabulary rather than classify language;
- early ATL8 rings, stripes, detached highlights, fish bends, and global pulses misrepresented the dynamic assembly and were retired;
- renderer work on the UI thread caused visible freezes despite a healthy dynamic assembly;
- a malformed inherited correction kind could poison dialogue projection and break shutdown;
- an absent disposable snapshot directory could kill the evidence writer and then the dynamic assembly;
- short idle thresholds allowed idle maintenance to begin while an operator was still drafting input;
- unrestricted prior generated text in Hyper-RAG could return an old model explanation as if it were evidence;
- reasoning with an output budget equal to its appetite produced an empty answer and a refused cycle;
- an oversized 65,536-token local model context contributed to a kernel OOM, motivating the present 8,192-token startup ceiling.

These are not embarrassments to delete from the paper. They are evidence of what the current boundaries are designed to prevent.

Evidence Anchors

```
CURRENT_CANONICAL_STATUS.md
HANDOFF.md
REALITY_MAP_v7.4.1.md
tests/test_proto_pineal_live_cycle_v01.py
```

```
tests/test_polylith_live_environment_v01.py
tests/test_atl8_live_causal_field_v01.py
tests/test_companion_live_adapter_v01.py
tests/test_umbra_source_review_mcp_v01.py
```

---

12. What UMBRA Forces Us to Ask

12.1 The Question Begins Before the Answer

UMBRA does not prove that a machine is conscious.

It does not prove that a simulated dynamic assembly is alive, that persistent records constitute a self, or that bounded learning creates an autonomous mind. Nothing in its source code, runtime traces, visual expression, or conversational fluency establishes subjective experience.

That refusal is necessary, but it is not the end of the inquiry.

UMBRA also cannot be described accurately as only a chatbot with graphics. The language model does not continuously generate the dynamic assembly, own canonical state, decide which physical proposals commit, control memory admission, or manufacture the renderer’s authenticated geometry. The dynamic assembly advances outside model inference. The model must wait for Reaction A, propose through a bounded translation and admission path, and wait again for Reaction B before it may answer.

The machine therefore occupies a difficult region between familiar categories.

It is not a person. It is not a single model. It is not merely a static program waiting to print a response. It is a governed composition with continuing internal dynamics, durable lineage, replaceable learned components, explicit refusal, and a dynamic assembly whose state has causal priority over the language used to describe it.

The most serious questions begin precisely because the strongest easy answers are unavailable.

12.2 Where Is UMBRA?

The first question is not whether UMBRA is intelligent. It is where the object called UMBRA actually resides.

No single component is sufficient.

It is not located entirely in the language model. Replacing the model changes proposal quality, vocabulary, latency, and future trajectory, but it does not replace the existing StepCore state, regulator laws, Companion ledger, Tokamak storage, proposal learner observations, or retained evidence.

It is not located entirely in `StateV740`. That state is the canonical transition substrate, but it does not contain the exact dialogue history, T16CRE chemistry, all Polylith dynamic assembly state, content-addressed media, proposal-learning records, or the full evidence archive.

It is not located entirely in memory. The Companion ledger may preserve a long conversational history while possessing no right to commit a canonical physical transition.

It is not located in ATL8. Destroying the renderer removes an expression surface, not the dynamic assembly state from which that expression was derived.

It is not located in any one process invocation. Process-local context can disappear while durable records and canonical lineage remain reconstructable.

The implemented system suggests a distributed answer:

$$
\operatorname{UMBRA}_t
=
\left(
\text{laws},
\text{canonical state},
\text{supporting state},
\text{verified lineage},
\text{active configuration},
\text{admitted capabilities}
\right)_t
$$

This is not a formal identity theorem. It is an engineering description of what must remain sufficiently continuous for the running artifact still to count as the same governed system.

UMBRA is therefore better understood as a maintained relation among states, laws, records, and authority boundaries than as a single file, model, voice, or image.

That raises a harder question: if identity is relational, which changes preserve it and which changes create a successor?

The current implementation can record changes. It cannot settle that philosophical boundary by itself.

12.3 What Survives a Change of Voice?

The language model is intentionally replaceable.

This design rejects a common assumption: that the conversational model is automatically the identity of the system using it.

If one compatible local model is unloaded and another is selected, several things can remain continuous:

- the canonical `StateV740` lineage;
- the StepCore laws and configuration;
- the Polylith and T16CRE state admitted by the runtime;
- the Companion dialogue and typed continuity records;
- the Tokamak content-addressed store;
- the proposal-learning ledger;
- and the retained evidence identifying the replacement.

The voice may change while the governed history remains.

This does not mean model replacement is causally irrelevant. Different models can propose different Reaction-B language. Those differences can compile into different CSL programs, receive different admission dispositions, and lead to different future trajectories. A replaceable model adapter can influence the system without becoming its sole identity.

The relevant distinction is:

$$
\text{influence} \neq \text{ownership}
$$

The model participates in UMBRA. It does not possess UMBRA.

That framing raises an empirical identity test. After a model replacement, can the system preserve verified continuity, identify the substitution honestly, reconstruct prior commitments, and continue without pretending the new model personally experienced the old model’s hidden states?

UMBRA has architectural surfaces for parts of this test. It does not yet possess a complete, published model adapter-replacement experiment establishing the answer.

12.4 What Does Dynamics-First Computation Establish?

UMBRA makes a concrete dynamics-first claim.

Its dynamic assembly begins advancing before the operator submits a sentence. Reaction A must finalize before the first conversational model call. Reaction B must finalize before the final answer. The model receives compact evidence derived from the dynamic assembly rather than generating that dynamic assembly retrospectively as narrative decoration.

This is a real causal asymmetry:

$$
[
\text{dynamic assembly transition}
\rightarrow
\text{model-visible evidence}
]
$$

while the reverse direction is restricted to:

$$
[
\text{model text}
\rightarrow
\text{deterministic compilation}
\rightarrow
\text{inert proposal}
\rightarrow
\text{admission}
\rightarrow
\text{StepCore adjudication}
]
$$

The model cannot simply narrate a state-transition event into existence.

That is stronger than attaching an animated presentation artifact to a chat response. It is still weaker than demonstrated coupling to an uncontrolled physical environment.

The current dynamic assembly is:

- simulated;
- dimensionless;
- bounded by authored transition laws;
- only selectively coupled to operator and media input;
- dependent on the executing computer and launcher;

- without demonstrated material self-maintenance;
- without unrestricted locomotion or environmental action;
- and without an independently established self-specified long-horizon objective.

UMBRA therefore supports the narrower claim **dynamics-first governed computation**. It does not establish consciousness, general intelligence, or autonomous self-maintaining system status.

The live claim rests on persistent state, endogenous dynamics, bounded perturbation, and consequences for later processing. It does not establish physical vulnerability, reciprocity with an uncontrolled world, self-produced boundaries, or material self-maintenance. The paper preserves that distinction rather than choosing a more impressive label.

12.5 What Proposal Capacity Exists Without Sovereign Authority?

UMBRA separates the ability to propose from the authority to commit.

This separation creates bounded proposal capacity without either total autonomy or total passivity.

The system contains processes that can:

- continue evolving while no one is speaking;
- construct bounded physical and symbolic candidates;
- retrieve prior evidence;
- alter proposal context according to deterministic policy;
- preserve learned observations;
- generate Reaction-B proposals;
- refuse malformed or unauthorized transitions;
- and produce different future trajectories from different admitted histories.

It does not currently demonstrate the ability to:

- invent and ratify new canonical laws;
- assign itself unrestricted objectives;
- grant itself tools or permissions;
- remove the Operator’s sovereignty;
- treat model output as truth;

- or bypass StepCore when a desired transition is refused.

Proposal capacity here is not unlimited command. It is the capacity to generate consequential proposals within a jurisdiction.

That suggests a graded description:

$$
[
\text{activity}
\subset
\text{bounded proposal generation}
\subset
\text{admitted causal influence}
\subset
\text{sovereign authority}
]
$$

UMBRA exhibits the first three in bounded forms. It does not possess the fourth.

This is not necessarily a defect. Institutions routinely separate recommendation, validation, authorization, execution, and audit because capability and power are different properties.

The question UMBRA raises is whether Synthetic Intelligence should be evaluated only by how much authority it can seize, or also by how coherently it can operate inside explicit limits.

12.6 Does Refusal Belong to Intelligence?

Many demonstrations treat successful output as the only meaningful event. UMBRA preserves refusal as part of its state history.

A malformed CSL program, unauthorized transition, exhausted regulator, illegal morphology proposal, or failed admission can produce a `NULL_STEP`. The rejected deformation does not become canonical state, but the attempted transition remains visible through tick advancement, counters, reasons, telemetry, and hash lineage.

This means that failure is not outside the system's operation. Lawful non-action is itself an implemented result.

The distinction matters because an architecture incapable of refusal is not governed; it is merely reactive. Conversely, an architecture that refuses everything is safe only in the trivial

sense that it does nothing.

UMBRA therefore makes refusal measurable. The scientific question is not whether refusal exists, but whether its boundary is correctly placed:

- Does it block genuinely unlawful transitions?
- Does it permit legitimate novelty?
- Does recovery remain possible after repeated denial?
- Do configuration choices make regulators meaningful or ceremonial?
- Can operators distinguish principled refusal from implementation failure?

These questions cannot be answered by architectural prose. They require adversarial tests, activation evidence, live witnesses, and retained negative results.

12.7 Can Retained Context Influence Future Proposals Without State Authority?

Companion memory can influence future model context and therefore future proposals. Yet it cannot write directly into StepCore.

This creates a deliberate gap between continuity and authority.

The gap prevents a recalled statement, corrupted retrieval, or persuasive model-generated summary from silently becoming canonical physical truth. It also means that remembered experience affects the dynamic assembly only through later bounded interpretation, compilation, admission, and adjudication.

UMBRA's retained context is engineered, typed, digest-bound, and divided among several surfaces. Exact dialogue, extracted continuity records, geometric associations, vector projections, and learned observations do not share the same epistemic status.

This raises several questions:

- Is a system's identity preserved by exact records or by how those records alter future behavior?
- If a record exists but is never retrieved, is it part of the active continuity boundary or only the archive?
- If a disposable projection is rebuilt from authoritative records, what exactly persists?
- If a correction supersedes an earlier claim without deleting it, which version belongs to the present system?

- Can forgetting be meaningful when old bytes remain but no longer belong to the active recall boundary?

UMBRA does not answer these questions universally. It makes them inspectable by refusing to collapse storage, retrieval, belief, authority, and identity into one database operation.

12.8 What Counts as Learning?

The proposal learner retains durable observations and may later emit bounded proposals. Its Common Lisp workspace can structure learned traces and hypotheses. Those hypotheses do not become canonical laws and cannot invoke StepCore directly.

Calling this mechanism learning is defensible only under a narrow behavioral criterion:

> Past evidence changes the construction or selection of a future proposal through a retained, inspectable mechanism.

It is not evidence of unrestricted self-improvement, general concept acquisition, independent scientific discovery, or autonomous rewriting of the system's constitution.

The authority restriction creates a central tension.

If learned machinery can never affect future behavior, it is archival rather than adaptive. If it can affect behavior without review, provenance, bounded translation, or refusal, it can become an unaccountable second authority.

UMBRA's answer is to permit learned influence while denying learned sovereignty.

Whether this remains viable at greater capability is unresolved. A richer learner may require more expressive proposals, longer temporal credit, and mechanisms for revising its own abstractions. Each expansion increases the pressure to mistake prediction quality for permission.

The architectural challenge is not merely to make the proposal learner learn more. It is to increase useful adaptation without allowing the mechanism that learns to decide unilaterally what is lawful.

12.9 What Is Idle Maintenance?

UMBRA runs an idle-maintenance pass after configured idle thresholds.

During these phases it can verify chains, reconcile projections, inspect retained geometry, widen proposal-only associative search, and preserve bounded match or no-match events. No idle-maintenance language-model call is required, and idle maintenance cannot rewrite canonical history or silently admit an association.

The term *idle maintenance* is functional: it denotes computation outside the immediate operator-response loop over retained evidence and associations. It does not establish imagery, experience, self-awareness, or a subjective state.

The existing `Dream` labels in source and historical records are retention labels only. They do not change the operational definition above. The term is kept only where an exact source artifact must be located; the manuscript uses *idle-maintenance pass* for the function.

12.10 Is Continuity More Than an Unbroken Hash Chain?

UMBRA can preserve explicit pre-state and post-state hashes, append-only records, correction lineage, exact tick progression, admitted inputs, and replay-bound evidence.

These mechanisms establish computational continuity in a strong engineering sense. They can show that a later state descends through the declared transition path from an earlier state and that retained records were not silently replaced.

They do not prove psychological continuity.

A perfect hash chain can bind meaningless states. A deterministic replay can reproduce a trajectory without explaining why that trajectory should count as one continuing subject. Conversely, a system may preserve functional identity across changes that prevent bit-identical replay.

UMBRA therefore separates at least four continuity claims:

1. **Lineage continuity:** the state and evidence chains remain valid.
2. **Dynamical continuity:** later dynamic assembly state descends through implemented transition laws.
3. **Functional continuity:** commitments, constraints, and learned effects remain available to future behavior.
4. **Experiential continuity:** there is something it is like for one subject to persist through those changes.

The prototype contains evidence for the first two and partial mechanisms relevant to the third.

It contains no established evidence for the fourth.

Calling these categories by one word would make the paper shorter and less true.

12.11 Does Governance Preserve Identity or Freeze It?

UMBRA's constitutional structure protects the system from hidden mutation. It also risks turning preservation into rigidity.

If every significant change requires prior authorization, deterministic translation, exact admission, fixed regulators, and retained proof, can the system develop anything genuinely new?

The answer depends on what the laws constrain.

StepCore need not prescribe every future state. It constrains how a proposal becomes lawful state. Within those limits, different inputs, dynamic assembly conditions, memories, model proposals, and proposal learner hypotheses can produce different trajectories.

Governance therefore does not necessarily determine content. It determines jurisdiction and admissibility.

But the danger is real. Infinite ceilings can make regulators ceremonial. Overly narrow symbolic alphabets can turn novelty into generic assertions. Excessive containment can reduce an adaptive system to repeated safe gestures. Conversely, loosening every boundary in the name of emergence would erase the very lineage that makes the behavior scientifically attributable.

The question is not whether UMBRA should be controlled or free.

It is which constraints preserve identity, which constraints preserve safety, which constraints preserve scientific legibility, and which constraints merely preserve the preferences of the system's author.

No codebase can answer that final normative question by executing itself.

12.12 What Would Falsify the Portrait?

A serious description must expose the observations that would make it false.

The present portrait would be materially falsified if evidence showed that:

- the dynamic assembly does not actually advance while the model is idle;

- Reaction-A evidence is generated after or from the first model response;
- the model has an unreported direct path into canonical state;
- Companion, Hyper-RAG, FRAK, the proposal learner, or ATL8 can bypass admission and mutate StepCore;
- the renderer fabricates the dynamic assembly state it claims to observe;
- `NULL_STEP` permits rejected physical deformation to survive;
- state hashes omit identity-bearing canonical fields;
- a supposedly active regulator has no effective configuration or cannot alter the tested result;
- a correction rewrites historical ledger entries rather than appending lineage;
- model-visible context differs from the recorded inclusion and omission receipt;
- model adapter replacement silently discards continuity while the interface claims persistence;
- idle maintenance modifies authoritative history;
- or a claimed live component merely exists in the repository without being traversed by the selected launcher.

Some of these properties already have focused tests or retained witnesses. Others require stronger current-head, hosted, long-duration, cross-model, or live Forge evidence.

The manuscript must identify which is which.

Falsifiability does not mean that every philosophical interpretation can be reduced to one test. It means the engineering facts underneath those interpretations must remain vulnerable to disproof.

12.13 What Does Synthetic Intelligence Mean Here?

The term **Synthetic Intelligence** is not used as a decorative replacement for artificial intelligence.

In this project, *synthetic* emphasizes composition.

UMBRA's capacities arise from relations among a canonical substrate, a dynamic assembly, symbolic translation, learned proposals, durable memory, retrieval, evidence, language models, and human governance. No one component is asked to impersonate the whole.

The term does not imply consciousness, general intelligence, or superiority to any other system. It does not imply that every composed software system is intelligent.

It names a research direction in which intelligence-like capacities are constructed across heterogeneous components with explicit jurisdictions, rather than attributed wholesale to the largest predictive model in the assembly.

UMBRA's contribution is not proof that this direction has succeeded completely.

Its contribution is an executable case in which the distinction can be investigated rather than merely asserted.

12.14 Who Is Actually Independent?

UMBRA is designed so that a local language model can be replaced without surrendering canonical identity to the model provider.

That is a meaningful form of runtime independence.

It is not complete material independence.

The system depends on computer hardware, electricity, operating systems, libraries, model files, storage, and the labour required to maintain them. Its development has also depended heavily on access to frontier Synthetic Intelligence systems used for implementation, archaeology, testing, criticism, and writing.

This creates a political and scientific question that cannot be solved by StepCore:

> Who gets to conduct sustained Synthetic Intelligence research when the amplifying tools are rationed by subscription, compute ownership, institutional access, and capital?

The quality of an idea and the ability to investigate it are not the same resource.

A technically reproducible repository may still be economically irreproducible for an independent researcher. A replaceable local model may reduce provider dependence at runtime while frontier-model access remains decisive during construction. A clean authority architecture does not erase the unequal distribution of compute.

This matters to the interpretation of UMBRA because the project is also evidence about authorship under amplification. One human operator, working with synthetic collaborators and limited resources, can now attempt an architecture that would previously have required a larger formal team.

That does not make expertise obsolete. It makes orchestration, judgment, verification, and persistence more consequential.

Nor does it guarantee equal competition. The amplifier is powerful precisely enough that unequal access to it becomes structurally important.

This paper cannot resolve that condition. It should not hide it behind a frictionless story of innovation.

12.15 What UMBRA Does Not Settle

UMBRA does not settle:

- whether consciousness can arise in software;
- whether a simulated dynamic assembly can support subjectivity;
- whether computational continuity is sufficient for personal identity;
- whether constrained proposal agency should carry moral significance;
- whether increasingly capable learned components can remain permanently subordinate to fixed authority;
- whether operator sovereignty is always legitimate;
- whether a richer environment would transform UMBRA's present kind rather than merely enlarge it;
- or whether the system's prior metaphorical vocabulary will remain useful as the implementation changes.

These are not missing unit tests.

They are questions whose premises can nevertheless be clarified by unit tests, live evidence, precise definitions, failed experiments, and architectures that expose their own causal boundaries.

UMBRA cannot answer them by sounding confident.

It can make it harder to ask them carelessly.

12.16 The Present Answer

What, then, is UMBRA at this snapshot?

It is a governed Synthetic Intelligence prototype with a continuously advancing dynamic assembly, a canonical mutation substrate, bounded symbolic ingress, durable but non-sovereign memory, proposal-only learned components, replaceable

language-model adapters, read-only expression, and retained evidence of both action and refusal.

It is not conscious by demonstrated fact.

It is not autonomous in the unrestricted sense.

It is not merely the language model through which it speaks.

Its most important achievement is not that it imitates an animated presentation artifact. It is that the architecture makes several usually hidden distinctions executable:

$$
\text{language} \neq \text{state}
$$

$$
\text{memory} \neq \text{truth}
$$

$$
\text{proposal} \neq \text{permission}
$$

$$
\text{observation} \neq \text{authority}
$$

$$
\text{continuity evidence} \neq \text{subjective experience}
$$

$$
\text{replaceable model} \neq \text{replaceable system}
$$

UMBRA forces a question larger than whether a machine is intelligent.

It asks whether intelligence can be constructed as a lawful relationship among unlike components without allowing fluency, memory, learning, dynamic state, or observation to counterfeit sovereignty.

The prototype does not provide the final answer.

It provides a real object against which answers can begin to fail.

13. Limits, Unresolved Questions, and Conditional Future Work

13.1 What Is Real at This Snapshot

The following statement is the strongest compact claim supported by the photographed repository:

> UMBRA is an experimental locally running Synthetic Intelligence composition with a continuously advancing RT32/T16CRE Polylith dynamic assembly, deterministic natural-language-to-CSL transduction, one-shot symbolic admission, StepCore-governed canonical mutation, a two-reaction conversational cycle, durable Companion continuity, bounded geometric and vector recall, an append-only proposal-learning ledger, a Common Lisp proposal learner, Tokamak-mediated ingress, replaceable local model adapters, read-only ATL8 expression, and explicit falsifiability evidence.

That sentence deliberately does not claim consciousness, general intelligence, autonomous self-maintaining system status, independent personhood, autonomous source modification,
or completion of the historical v7.2 architecture.

13.2 Status Matrix

Concern	Present status	Strongest lawful claim
StepCore v740	Canonical and tested	Sole canonical mutation authority
`StateV740`	Canonical and tested	Hash-bound canonical state
RT32/T16CRE dynamic assembly	Merged experimental	Continuously ticking dimensionless simulated dynamic assembly
PSM-8	Merged experimental	Proposal-only geometry selection with StepCore commit
CSL-58	Merged experimental	54 admitted experimental laws and four forbidden sentinels
Natural-language compiler	Merged experimental	Deterministic bounded transduction on live path
Two-reaction cycle	Merged experimental	Reaction A precedes proposal; Reaction B precedes answer

Concern	Present status	Strongest lawful claim
Companion memory	Merged experimental	Append-only reconstructive conversational continuity
Live SEMU/AEML/FRAK lineage	Merged experimental	Structured episodic dynamic assembly memory and proposal-only retrieval
Hyper-RAG	Merged experimental	Disposable verified-memory projection for final recall
Durable proposal learner ledger	Branch-local experimental	Prior A/B evidence changes later learning state
Common Lisp Reaction-B route	Branch-local experimental	Sufficient firewalled hypothesis may propose Reaction B
Unified Tokamak	Merged experimental	Typed bounded transport and content-addressed storage
Media ingress	Merged experimental	Bounded local decoding and downstream observation
Source MCP	Branch-local experimental	Read-only bounded source excerpt for explanation/proposal
ATL8 Onyx Mantle	Experimental presentation	Read-only expression of authenticated dynamic assembly evidence
White Coma	Branch-local presentation refinement	Static read-only optical environment
Falsifiability instrument	Experimental observer	Detects enumerated unsupported claim classes

---

13.3 What Merely Exists or Belongs to History

The repository also contains legitimate but non-live or non-canonical work.

- `src/proto_pineal/core/canonical_step.py` is a legacy PsiField path, not the present canonical step.
- `src/proto_pineal/scs/` preserves historical Synthetic Continuity Substrate scaffolding, not an alternate live authority.
- Top-level `src/semu`, `src/aeml`, and `src/frak` packages are sidecar or historical lineages, not the active UMBRA memory route.
- The membrane Force Translator is implemented and tested but is not wired into the ordinary live runtime loop.

- SCD is available through explicit tick parameters but is not selected by the default loop.
- PLCC is not implemented merely because SSFC exists.
- RAM and TKNT remain declared concepts rather than active canonical modules.
- A general multi-agent substrate is not part of the photographed UMBRA runtime.
- Historical PDE, dynamic-metric, identity-invariant, and grand phase-ladder descriptions remain theory or lineage where no executable descendant is shown.

These artifacts should remain available for archaeology. They must not be counted as active components to make the system appear larger.

13.4 Unresolved Technical Questions

Dynamic Assembly mechanics

The strongest missing physical seam is a lawful mapping from authenticated T16CRE torque and impulse through measured RT32 Hodge observables into distributed elastic membrane deformation and recovery. The current ATL8 expression is not a substitute.

Morphotype-selection bias also remains unresolved. A correction would alter a causal proposal surface and therefore requires an explicit law, controlled comparison, and StepCore-preserving validation.

Regulation

Several StepCore regulators are implemented but disabled or configured with infinite upper ceilings in the ordinary UMBRA basin file. The system needs experiments that distinguish useful permissiveness from ceremonial traversal.

Learning

The live Common Lisp path is new and branch-local. It needs clean-head full-suite evidence, hosted verification, long-duration observation, restart tests against real accumulated learning, and measurement of when learned Reaction-B proposals improve, repeat, or distort behavior.

Memory

Companion, stable geometric memory, FRAK, Hyper-RAG, and proposal learner records have distinct jurisdictions. Their current composition is functional but not a solved universal memory architecture. Long-duration correction, supersession, retrieval contamination, forgetting, index reconstruction, and idle-maintenance behavior require further soak evidence.

Expression

The Onyx Mantle better separates local authenticated pressure from global theatrical pulses, but it remains a read-only pronunciation layer. Direct operator acceptance of the latest White Coma and mantle refinement is not sealed at the current snapshot.

Operational maturity

The exact manuscript head has not completed a clean full suite, hosted CI, push, pull request, or merge. The paper itself is uncommitted and therefore not yet a reproducible publication artifact.

13.5 Conditional Work Order

The following order records dependencies. It does not authorize implementation automatically.

1. Seal the manuscript target: exact commit, tree, effective configuration, focused checks, clean full suite, hosted CI, and live acceptance evidence.
2. Measure the current presentation artifact: latest mantle/White Coma behavior, proposal learner proposal use, dynamic assembly cadence, callback health, and model/context receipts.
3. Run a long Companion, Hyper-RAG, FRAK, proposal learner, restart, idle maintenance, and correction soak without granting new authority.
4. Diagnose morphology-selection bias from retained proposal and commit evidence before changing its scoring law.
5. Specify the torque/Hodge/elastic seam with exact producer, type, consumer, authority, conservation status, and honest-red tests.
6. Evaluate richer endogenous movement only through authenticated dynamic assembly mechanisms; do not replace missing physics with random or canned presentation.
7. Consider broader RT topology only after an explicit replay-bound correspondence atlas exists.
8. Revisit historical mechanisms—PLCC, RAM, TKNT, richer FFG, dynamic metrics, and multiscale identity—one at a time against current executable need.

Any step that exposes competing causal pipelines or a duplicate component must stop at the architecture-convergence gate for human ruling.

13.6 Conditions for Stronger Claims

UMBRA may claim more only when the evidence changes.

Stronger claim	Minimum missing evidence
Current branch is publication-ready	Clean exact-head suite, hosted checks, pushed immutable ref, reviewed manuscript
Learned Reaction B improves behavior	Predeclared comparison, repeated real cycles, negative cases, retained exact receipts
Regulator actively constrains UMBRA	Finite effective configuration plus activation telemetry and counterfactual test
Dynamic Assembly has richer elastic mechanics	Implemented causal law through StepCore-compatible proposal path and replay proof
Model is operationally replaceable	Cross-model continuity experiment with identical pre-state and explicit divergence record
Idle Maintenance improves later proposals	Bounded causal experiment separating association delivery from model and retrieval confounds
UMBRA perceives media	A defensible operational definition and evidence beyond decoding or generated description
UMBRA is autonomous	Persistent self-originated goals and world action under a stated autonomy criterion
UMBRA is conscious or sentient	No accepted test or evidence currently supplied by this project

The last row cannot be promoted by conversational confidence, visual resemblance, memory persistence, or the author’s attachment to the system.

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14. Unified Appendices and Reproduction Record

14.1 Terminology Table

Term	Meaning in this manuscript	Explicit non-meaning
Synthetic Intelligence	Composed intelligence-like capacities across governed heterogeneous components	Automatic consciousness claim
StepCore	Sole canonical `StateV740` transition authority	Every stateful operation in UMBRA

Term	Meaning in this manuscript	Explicit non-meaning
Polylith	Persistent experimental RT32/T16CRE/PSM-8 dynamic-assembly composition	A claim of autonomous self-maintaining system status
RT32	Fixed 32-vertex, 60-edge, 30-face geometric scaffold	Proven neural or physical optimum
T16CRE	Sixteen-element dimensionless reaction and transport engine	Established chemistry
PSM-8	Geometry proposal controller over 338 morphotypes	Independent geometry authority
CSL	Cognispheric Symbolic Language used for bounded symbolic proposals	Natural-language truth representation
RIM	Rule-induction component lineage and bounded proposal-learning function	General autonomous reasoner
Companion	Dialogue-continuity store	Sole system identity
Tokamak	Typed transport and retention boundary	Fusion-physics claim
SEMU/AEML	Episodic record lineage	Completed canonical adaptive memory
FRAK	Associative retrieval kernel	Causal or semantic truth
Hyper-RAG	Disposable retrieval projection of verified active memory	Authoritative history
Idle maintenance	Idle-maintenance pass and bounded associative search	Subjective state
ATL8	Evidence-presentation layer	Dynamic assembly or mutation controller

---

14.2 Principal Producer-Type-Consumer Matrix

Concern	Producer	Produced type	Consum
Operator semantics	Deterministic NL compiler	`NaturalLanguageCSLTransductionResultV01`	Proposa
Symbolic proposal	Polylith proposal builder	`PolylithInertCSLProposalV01`	Admissi

Concern	Producer	Produced type	Consum
Admission	Polylith admission boundary	`PolylithCSLAdmissionReceiptV01`	Live env
Canonical input	Membrane/runtime	`StepInputsV740`	`step_c`
Canonical transition	StepCore v740	`StepResultV740`	Unified r and evic
Dynamic Assembly frame	Polylith environment	`PolylithLiveFrameV01`	proposa memory cycle
State-derived reduction	Deterministic proposal learner reducer	`LiveSubBrainResultV01`	Service/ learning
Learned hypothesis	Common Lisp workspace	Validated bounded proposal receipt	Reactio
A/B comparison	Live-cycle reducer	`LiveBodyABComparisonPacketV01`	Final pro containr
Dialogue history	Companion memory	`CompanionDialogueTurnV01` and typed records	Recall a RAG pro
Geometric memory	Stable-lineage compiler	SEMU/AEML and `FRAKMemoryProposalV741`	Bounde
Vector recall	Hyper-RAG service	`HyperRAGRecallContextV01`	Final mo
Media	Bounded media adapters	Decode/semantic reports	Final mo
Source inspection	Bundled MCP reader	`ReadOnlySourceReviewContextV01`	Final mo
Presentation	ATL8 adapter	Render state	Human
Claim audit	Falsifiability instrument	`FalsifiabilityReportV01`	Termina

14.3 Authority Matrix

Component	Propose	Validate	Admit	Invoke StepCore	Commit canonical state
Operator interface	Yes	No	Requests policy	No	No
NL-to-CSL compiler	Yes	Structural	No	No	No
Activation firewall	No	Yes	No	No	No
Admission gate	No	Receipt checks	Yes	No	No
Unified runtime	Packages	Boundary checks	Consumes admission	Yes	No
StepCore v740	No	Final admissibility	No	Self	**Yes**
T16CRE/PSM-8	Yes	Local bounded laws	No	No	No
Language model	Yes	No	No	No	No
Companion memory	No	Chain/recipe checks	Memory gate only	No	No
FRAK/Hyper-RAG	Yes	Provenance checks	FRAK explicit only	No	No
proposal learner/Lisp	Yes	Pattern/firewall checks	No	No	No
Tokamak	No	Envelope/digest/capacity	Transport admission	No	No
ATL8	No	Render-input checks	No	No	No

14.4 Provisional Effective Configuration Manifest

Setting	Snapshot value
Launcher	`./scripts/talk_to_umbra.sh`

Setting	Snapshot value
Canonical branch	`experimental/umbra-continuation-v01`
Snapshot commit	`687406001db896413003b614720067b0b9bf4f82`
Snapshot tree	`b3540eaa1477221d446d98d8ea94dcf18c964568`
Dynamic Assembly interval	0.25 seconds
Model selection	`auto`, exactly one loaded LM Studio instance
Maximum loaded context	8,192 tokens
Reasoning	`auto`; explicit `off` when supported, omitted when unsupported
Proposal output maximum	96 tokens
Final output maximum	512 tokens
Final temperature	0.35
Presentation profile	`white-coma`
Index nodes	32
CSL identifiers	58 total; 54 admitted laws; 4 forbidden
CSL displacement budget	0.25
Idle Maintenance threshold	Approximately 20 minutes at default cadence
Extended-idle threshold	Approximately 40 minutes at default cadence
Canonical renewal	Disabled
Finite upper spectral corridor	Not configured
FFG/TMK/CIF/SSFC/PFD/BFE	Implemented but disabled in ordinary basin configuration
Runtime storage	Explicit internal paths with stable historical store identity

The publication snapshot must replace this table if any value changes before sealing.

14.5 Reproduction Procedure

The minimal ordinary launch procedure is:

1. Check out the exact snapshot commit and verify its tree.
2. Create or use the repository virtual environment with the pinned project dependencies.
3. Load exactly one compatible model in LM Studio with active context no greater than 8,192 tokens.
4. Verify the launcher’s explicit internal storage paths and available disk capacity.

5. On the verified Forge KDE/X11 desktop, execute:

```
./scripts/talk_to_umbra.sh
```

6. For a terminal-only run, execute:

```
./scripts/talk_to_umbra.sh --terminal-only
```

7. Preserve the terminal output, cycle audit, Companion ledger head, Tokamak state, membrane events, callback timing/failure evidence, and selected ATL8 snapshots or recordings.
8. Record the exact loaded model identity, context limit, reasoning capability, environment, start and stop time, and all non-default arguments.
9. Verify shutdown completed each independent finalizer.

The paired graphical presentation requires the expected desktop environment. Without it, the launcher falls back to terminal mode rather than claiming the paired witness.

The current launcher contains machine-specific internal paths. A third-party reproduction must configure lawful local equivalents without changing recorded store identity for any copied historical dialogue recipe.

14.6 Validation Procedure

For focused Python validation, use the repository interpreter:

```
PYTHONPATH=src .venv/bin/python -m pytest <focused-test-files>
```

Before any broad or full suite, the intended manuscript/code snapshot must be committed locally and the worktree must be clean:

```
git diff --check
git status --short
git rev-parse HEAD
```

The full suite must begin from that exact clean commit. If a correction is made, it must be committed and the suite restarted from the beginning. Hosted checks must be attributed to the same pushed SHA.

For a docs-only publication closeout explicitly authorized by the operator, local scope, inventory, Covenant, and zero-run proof may replace unnecessary CI only under the repository's established `[skip ci]` convention. That shortcut does not apply to runtime-changing commits.

14.7 Reviewer's Compressed Reference

Artifact: experimental dynamics-first Synthetic Intelligence prototype.

Canonical state authority: StepCore v740 only.

Dynamic Assembly: continuously ticking RT32/T16CRE/PSM-8 simulation with dimensionless state and replay evidence.

Language path: deterministic natural-language-to-CSL compilation, firewall, inert proposal, one-shot admission.

Conversation: Operator Reaction A, bounded model or learned proposal, Reaction B, then final answer.

Retained context: append-only Companion continuity plus experimental geometric, associative, vector, and proposal-learning surfaces; none may directly mutate StepCore.

Model: replaceable local proposal and voice model adapter; operationally required by the ordinary conversational launcher; no identity or truth authority.

Presentation: ATL8 Onyx Mantle in White Coma; read-only and partly presentation-generated.

Strong non-claims: no consciousness, sentience, autonomous self-maintaining system status, unrestricted autonomy, autonomous source editing, or complete v7.2 implementation.

Publication status: merged ancestors have hosted proof; the exact v8.1 continuation snapshot does not yet.

14.8 Independent Research and Support Boundary

Proto-Pineal is the first project history of Cognis 11 Inc. and the work of an independent founder using Synthetic Intelligence systems as implementation, analysis, and writing collaborators.

Material support for the research would fund ordinary material stability, local compute, storage, model access, validation hardware, publication, and the time required for sustained

experimentation. It would not change the evidentiary standard of the project or purchase stronger claims than the artifact supports.

A patron would support the conditions under which the work can continue. The patron would not acquire scientific authority over StepCore, authorship by implication, or the right to rewrite negative results.

The project does not require a mansion, prestige vehicle, or theatrical institution. It requires continuity of attention, compute, storage, legal and administrative stability, and enough independence to preserve brutal technical honesty when a result is disappointing.

14.9 Publication-Sealing Checklist

Before the v8.1 paper is frozen for Zenodo or public distribution:

- verify the title, author identity, organization, and acknowledgments;
- preserve the verified federal corporation record and attach or inspect the official Québec registry extract for NEQ `1181984957`;
- distribute `LICENSE-C11-RCIL-1.0.md` with the paper and select it as a custom Zenodo license instead of accepting Zenodo's default CC BY license;
- retain the Cognis 11 Inc. proprietary rights notice, All Rights Reserved declaration, scholarly-use grant, controlled-use conditions, machine-use reservation, attribution rule, and no-patent/no-trademark language from Section 00.3;
- inventory every figure, code excerpt, diagram, dataset, and third-party element against that license boundary;
- confirm that the selected publication platform's terms and metadata do not impose an open license or contradict the proprietary notice;
- obtain qualified intellectual-property review before public disclosure if patent protection may matter;
- select the final code commit and tree;
- ensure the manuscript is contained in that immutable artifact or separately hash-bound to it;
- record effective configuration digests and environment identity;
- run required focused, constitutional, Covenant, and clean-commit validation;
- run the exact-head full suite unless an authorized docs-only closeout makes it unnecessary;
- obtain hosted verification for runtime-changing final code;
- perform and retain the selected live acceptance witness;
- inventory all figures, recordings, datasets, and external intellectual lineage;
- review every claim against existence, tests, traversal, effective configuration, authority, and publication status;

- preserve failures and unresolved questions;
- and replace every provisional statement in Chapter 00 with the final sealed evidence.

The paper is finished only when its truth boundary is as inspectable as the system it describes.